Bayesian X-Learner: Calibrated Posterior Inference for Heterogeneous Treatment Effects under Heavy-Tailed Outcomes

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Abstract

Conditional Average Treatment Effect (CATE) estimation in practice demands three properties simultaneously: heterogeneous effects $\tau(x)$, calibrated uncertainty over them, and robustness to the heavy tails that contaminate real outcome data. Meta-learners (Künzel et al., 2019) give (i); causal forests and BART give (i)–(ii) with Gaussian-tail assumptions; no widely used tool gives all three. We present *Bayesian X-Learner*, an X-Learner built on cross-fitted doubly robust pseudo-outcomes (Kennedy, 2020) with a full MCMC posterior over $\tau(x)$ via a Welsch redescending pseudo-likelihood. On Hill's IHDP benchmark the default configuration attains mean $\sqrt{\varepsilon_{\text{PEHE}}} = 0.56$ on 5 replications (lowest mean; differences from S-/T-/X-learners, full-config Causal BART, and a causal forest baseline are not significant at $\alpha = 0.05$, and rank ordering is unstable at 10 replications — IHDP comparisons are competitive rather than dominant). On contaminated "whale" DGPs with up to 20–25% tail density, a one-flag extension (`contamination_severity`) that selects a Huber-δ nuisance loss per Huber's minimax-δ relation recovers RMSE ≈ 0.13 with tight credible intervals (single-cross-fit 30-seed coverage 83% [Wilson 66%, 93%] at 20% density; modular-Bayes pooling with Bayesian-bootstrap nuisance draws restores nominal 95% coverage). We validate on the Hillstrom email-marketing RCT (N = 42,613), demonstrating consistent behaviour on real heavy-tailed outcome data, and report covariate-stratified $\tau(x)$ coverage across covariate quintiles to substantiate calibration for heterogeneous effects beyond scalar summaries. We draw a clean distinction between *tails-as-contamination* (handled by Welsch + Huber nuisance) and *tails-as-signal* (handled by a tail-aware CATE basis); an empirical probe confirms a tail-aware basis recovers τ_{tail} with full subgroup coverage, while the library's Hill-estimator path is contamination-directed and should not be used for heterogeneous τ. We map six empirical boundaries (contamination ceiling, clean-data efficiency cost, basis sensitivity, sample size, treatment type, compute) and show where other tools are preferable. Code and reproducible benchmarks are released.

1 Introduction

CATE estimation on real outcome data demands three properties that few tools jointly provide: heterogeneous $\tau(x)$, calibrated uncertainty on $\tau(x)$, and robustness to heavy tails. Meta-learners (Künzel et al., 2019) target the first; bootstrap CIs from causal forests (Wager & Athey, 2018) target the second approximately; classical Bayesian causal models (Hill, 2011; Hahn et al., 2020) target the second exactly but under Gaussian-tail assumptions that fail on medical outcome distributions with rare severe events, revenue distributions with long-tailed customer value, and response surfaces with structural heterogeneity. Real outcome data almost always has real tails, and Gaussian-likelihood Bayesian models systematically misfit them.

This paper introduces the *Bayesian X-Learner*: an X-Learner whose Phase-3 inference layer is a full MCMC posterior over $\tau(x)$ with a Welsch-form redescending pseudo-likelihood (Dennis Jr. & Welsch, 1978). Compared to Gaussian Bayesian baselines, the redescender trades a known efficiency cost on truly clean data

(Huber, 1964; Huber & Ronchetti, 2009) — the asymptotic relative efficiency for a univariate location estimand transfers, with finite-sample amplification we document empirically, to the CATE setting — for non-trivial robustness to heavy-tailed pseudo-outcomes. The resulting posterior is interpretable as a generalised Bayesian (Gibbs) posterior and supports the downstream tasks — subgroup comparisons, integrated decisions, propagated uncertainty — that point estimates with bootstrap CIs do not.

We distinguish two heavy-tail regimes the practitioner may face. *Tails-as-contamination*: outcomes contain a small fraction of units whose Y is distorted by nuisance mechanisms (measurement error, reporting artefacts, revenue whales in marketing) and whose true τ is nominal. *Tails-as-signal*: a minority subgroup has a genuinely different τ, and the practitioner cares about recovering that subgroup's effect. These regimes look superficially identical but call for opposite tools: the first wants downweighting (Welsch + Huber nuisance); the second wants representation (a tail-aware CATE basis). A controlled DGP probe (§5) confirms both pathways work as hypothesised, and that the library's data-layer `normalize_extremes` path is a contamination-reduction op that should not be activated for heterogeneous τ.

Our contributions are:

1. A Bayesian X-Learner that pairs doubly robust cross-fitted pseudo-outcomes (Kennedy, 2020) with a Welsch MCMC posterior over $\tau(x)$. On Hill's IHDP benchmark, the default configuration leads on mean $\sqrt{\varepsilon_{\text{PEHE}}} = 0.56$ against S-/T-/X- Learner, full-config Causal BART (Hill, 2011) at 0.60, and EconML's causal forest at 1.06, at the tightest dispersion among competitive entries.

2. An explicit separation of *tails-as-contamination* and *tails-as-signal* regimes, with a controlled probe (§5.5) and a graded basis-sensitivity ablation (§5.6) showing which library mechanism addresses which and why a data-layer Hill-estimator rescaling is actively harmful for the signal regime.

3. Empirical validation of Huber's 1964 minimax-δ prescription as an operational configuration: recovery at up to ∼20–25% whale density via a nuisance-loss selection exposed through a `contamination_severity` enum (§4.6), with documented breakdown beyond 30%.

4. Validation on the Hillstrom email-marketing RCT ($N = 42{,}613$) and covariate-stratified $\tau(x)$ credible band coverage across covariate quintiles (§5.10), substantiating calibration for heterogeneous effects beyond scalar ATE summaries.

5. Six mapped empirical boundaries (§6), each with a pointer to the benchmark that produced it.

Code, 17+ reproducible benchmarks, and this paper's source are released under an open-source licence; the repository URL is omitted from this submission and will be provided upon acceptance.

2 Background and Landscape

2.1 Position in the ML/AI landscape

Causal machine learning has moved from a statistical speciality to a first-class ML subfield. Uplift modelling in recommender systems (Wager & Athey, 2018), clinical decision support, and policy optimisation all require CATE estimators that produce *heterogeneous* effects rather than a single average. What purely predictive ML leaves on the table is calibrated uncertainty on those effects: a point estimate of "treatment lifts outcome by 3 units in subgroup A" is actionable only if we know whether the confidence band crosses zero and whether the band itself is trustworthy. Bayesian posteriors over $\tau(x)$ supply exactly this object — they support downstream operations (integration, propagation, decision rules) that point-and-bootstrap does not — but they are currently available only under Gaussian-tail assumptions. Our work targets the intersection: calibrated Bayesian $\tau(x)$ posteriors under heavy-tailed outcome data.

2.2 Use cases

Four grounded applications drive the library's design:

Tool	Heterogeneous $\tau(x)$	Calibrated uncertainty	Handles real tails
Classical S/T/X-Learner	✓	×	×
EconML Causal Forest	✓	bootstrap	×
Causal BART	✓	Bayesian	Gaussian only
TMLE / DML	partial	frequentist	×
Bayesian X-Learner	✓	✓(MCMC)	✓(Welsch)

Table 1: Tool coverage across the three properties practitioners need. Only the last row covers all three.

- **Clinical trials with rare severe outcomes.** Mortality events, adverse drug reactions, and rare efficacy spikes produce heavy-tailed outcome distributions. A robust posterior supports go/no-go decisions where bootstrap confidence intervals — which inherit the resampled distribution's tail sensitivity — are insufficient.
- **Marketing uplift with revenue whales.** Long-tailed customer value distributions are the norm, not the exception. Recovering both τ and calibrated uncertainty under whale contamination is what neither causal forests nor TMLE provide together.
- **Policy evaluation with structured $\tau(x)$ hypotheses.** When prior knowledge suggests τ depends on specific features (age, dose, covariate interaction), a user-supplied basis $\phi(x)$ encodes the hypothesis directly and returns a posterior over interpretable coefficients, not an opaque regression surface.
- **Research contexts where the full posterior matters.** Integrating over $\tau(x)$ to compute $\mathbb{P}(\tau(x) > \text{threshold})$, comparing posteriors across populations, or propagating uncertainty into downstream cost-benefit calculations. Point estimates plus bootstrap give the first; a Bayesian posterior gives all three from a single fit.

2.3 Problems with the current landscape

Classical meta-learners (Künzel et al., 2019) provide heterogeneous $\tau(x)$ but produce point estimates without uncertainty quantification. Causal forests (Wager & Athey, 2018) add bootstrap confidence intervals whose coverage on heavy-tailed outcome distributions has not been systematically validated. Causal BART (Hill, 2011) and BCF (Hahn et al., 2020) produce genuine Bayesian posteriors via Gaussian-process-inspired tree priors, but the likelihood is Gaussian: a single whale distorts the posterior through its squared residual. TMLE and double machine learning (DML, Nie & Wager, 2021) yield calibrated frequentist intervals for the average effect, with CATE extensions that remain an active research area; neither provides a Bayesian object for downstream decision-making under tails. The last row is the position this paper occupies: calibrated MCMC posteriors over $\tau(x)$ with a redescending pseudo-likelihood that tolerates real-world outcome tails.

3 Related Work

3.1 Meta-learners for CATE

Meta-learners decompose CATE estimation into a sequence of regression problems, each solved by an off-the-shelf supervised learner. The S-Learner fits a single outcome model $\mu(x, w)$ and takes $\tau(x) = \mu(x, 1) - \mu(x, 0)$; the T-Learner fits separate μ_0, μ_1 and takes their difference. Both are consistent but suffer from regularisation bias: the S-Learner under-fits τ when w is a weak predictor of y; the T-Learner over-fits each arm when one arm is much smaller than the other. The X-Learner of Künzel et al. (2019) repairs this by imputing counterfactual outcomes and regressing on the resulting pseudo-effects, then weighting the two fits by propensity. The construction is particularly well-suited to treated/control imbalance, which is the norm outside of randomised experiments.

Kennedy (2020) shows that replacing the X-Learner's imputed pseudo-outcomes with *doubly robust* (DR) pseudo-outcomes $D_1 = \mu_1 - \mu_0 + (Y - \mu_1)/\pi$ yields a target whose second-stage regression consistently

estimates $\tau(x)$ under a rate condition on the nuisance fits that is weaker than either the T- or X-Learner requires separately. The R-Learner of Nie & Wager (2021) takes a complementary quasi-oracle route via residualisation. Curth & van der Schaar (2021) survey the meta-learner landscape and formalise a theory of conditions under which each variant is preferred; a follow-up (Curth & van der Schaar, 2023) argues that much of the apparent model-selection variance in the CATE literature is benchmark-dependent rather than fundamental. Our method uses the DR pseudo-outcome target — the imputation and debiasing happen in Phase 2 of our pipeline (§4) — and inherits its rate properties. What classical meta-learners leave open is uncertainty quantification on $\tau(x)$: the standard recipe is bootstrap, which inherits the sensitivity of the base learner to outliers.

3.2 Bayesian causal inference and MCMC

Bayesian causal inference on $\tau(x)$ has been dominated by flexible tree-based priors. BART itself (Chipman et al., 2010) provides the backbone; Causal BART (Hill, 2011) adapts it to the potential-outcome framework, producing a posterior over the counterfactual response surfaces from which $\tau(x)$ is derived. BCF (Hahn et al., 2020) decouples the treatment effect from the prognostic surface using separate BART priors and yields substantial improvements on heterogeneous response; SBCF (Caron et al., 2022) extends BCF with horseshoe-style global- local shrinkage on the treatment-effect terms for sparse $\tau(x)$. Alternative Bayesian nonparametric formulations include Gaussian processes (Alaa & van der Schaar, 2018) and neural-network architectures with uncertainty estimation (Shi et al., 2019; Jesson et al., 2021). A recent line (Zheng et al., 2022) extends Bayesian CATE to multi-treatment and partial-identification settings. Frequentist semiparametric alternatives — generalised random forests (Athey et al., 2019), double/debiased machine learning (Chernozhukov et al., 2018), and conformal inference for individual treatment effects (Lei & Candès, 2021; Alaa et al., 2023; Jin et al., 2023) — deliver calibrated intervals through different routes (asymptotic normality, orthogonal scores, finite-sample distribution-free guarantees) but produce point-estimate-plus-interval objects rather than full posteriors over $\tau(x)$. Modern probabilistic-programming languages (NumPyro, Stan, PyMC) make Bayesian architectures easy to implement, and have been used to build custom CATE models with hierarchical priors or non-conjugate likelihoods.

The common thread is a Gaussian outcome likelihood on the target scale: either directly on Y (Causal BART) or on the transformed pseudo-outcome (most custom constructions). Under heavy tails this assumption distorts the posterior: a single whale's squared residual dominates the likelihood gradient, pulling the posterior mean toward the contaminated point and inflating the posterior variance. Our contribution is to replace the Gaussian layer with a Welsch redescending pseudo-likelihood while keeping the rest of the meta-learner + MCMC architecture standard.

3.3 Robust statistics for heavy-tailed regression

Huber (1964) introduced the minimax framework for point estimation under contaminated location models, proving that the Huber ψ-function $\psi_\delta(r) = r \cdot \mathbf{1}(|r| \le \delta) + \delta \cdot \mathrm{sign}(r) \cdot \mathbf{1}(|r| > \delta)$ is minimax-optimal over the ϵ-contamination neighbourhood $\mathcal{F}_\epsilon$. The optimal tuning constant δ is the unique solution of

$$\frac{\phi(\delta)}{\delta} - (1 - \Phi(\delta)) \;=\; \frac{\epsilon}{2(1-\epsilon)}, \tag{1}$$

where ϕ, Φ denote the standard normal PDF and CDF. Classical values are $\delta = 1.345$ at $\epsilon = 5\%$ and $\delta = 0.5$ at $\epsilon \approx 40\%$; the full table and a numerical derivation appear in Appendix B.3. The asymptotic relative efficiency (ARE) of Huber estimation vs maximum-likelihood under a truly Gaussian model follows from $\mathbb{E}[\psi_\delta^2]/\mathbb{E}[\psi_\delta']^2$ and is $\approx 95\%$ at $\delta = 1.345$ and $\approx 79\%$ at $\delta = 0.5$. Redescending M-estimators push robustness further by letting the influence function return smoothly to zero for very large residuals; the biweight (Beaton & Tukey, 1974) and the exponential Welsch form (Dennis Jr. & Welsch, 1978), $\psi_W(r) = r \exp(-r^2/2c^2)$, are the two canonical choices. We adopt Welsch for its smooth gradient and use it in the Bayesian likelihood. It gives up a small amount of additional efficiency for complete immunity to very large outliers. These tools are decades old and well understood as point estimators; their integration into Bayesian CATE posteriors is, to our knowledge, not standard practice.

3.4 Heavy-tailed inference for causal effects: five paradigms

Heavy-tailed causal inference is best understood through the *worldview* a method imposes on its tails. Five paradigms recur in the literature, each making a different concession.

(i) Robust M-estimation replaces the empirical mean with a concentration-friendly alternative — Catoni's mean (Catoni, 2012), median-of-means, or Huber-style M-estimators — while leaving the ATE/CATE estimand unchanged. The method becomes insensitive to extreme observations at a known efficiency cost on Gaussian data (Huber & Ronchetti, 2009); recent CATE-targeted variants (robust DR-learners, Chernozhukov et al. (2018)-style orthogonalised scores with robustified residuals) sit in this paradigm.

(ii) Weight stabilisation attacks the dual problem of heavy tails arising from the propensity side. Overlap weights (Li et al., 2018), trimming, and entropy-balancing mitigate IPW variance explosions either by capping weights or by redefining the target population so extreme weights vanish. Our `use_overlap=True` option implements the former.

(iii) Functional replacement changes the estimand itself — to a quantile, CVaR, or a full conditional counterfactual density (Kallus & Zhou, 2018). The tail is no longer a nuisance to be suppressed; it is the answer. Distributional and quantile treatment effects are the canonical form. Coverage of very extreme quantiles is bounded by empirical support, where extreme value theory (EVT; Beirlant et al., 2006) would extrapolate.

(iv) Distribution-free intervals. Conformal prediction applied to ITEs (Lei & Candès, 2021; Alaa et al., 2023; Jin et al., 2023) delivers finite-sample-valid prediction intervals without a tail model. They cannot extrapolate beyond observed support, and the recently developed sensitivity-aware variant of Jin et al. (2023) additionally relaxes unconfoundedness.

(v) Partial identification concedes that the target is not point-identified and reports sharp bounds (Yadlowsky et al., 2022). Robust to arbitrarily heavy tails within bounded-outcome assumptions, at the cost of producing a range rather than a number.

Where this paper sits. The Bayesian X-Learner is in paradigm (i) — it robustifies the estimator while keeping the ATE/CATE target intact — but adds two moves the standard M-estimation toolkit does not. First, it returns a *full Bayesian posterior* over $\tau(x)$ rather than a point estimate plus sandwich SE; calibration of credible intervals under contamination (§5.7) is the empirical justification. Second, it makes the *tails-as-contamination vs tails-as-signal* distinction operational: the same library handles paradigm-(i) contamination via Welsch + Huber nuisance, and tail-heterogeneous τ via a user-supplied basis $\phi(x)$ at the CATE-surface layer.

Connection to robust ATE pipelines. Industry pipelines for robust ATE estimation under heavy-tailed revenue outcomes have used trimmed and other bounded-influence estimators in place of plain difference-of-means or naive DR; for example, the Trimmed Match estimator of Chen & Au (2022) delivers a distribution-free, robust ATE for randomised paired geo experiments. These approaches typically terminate at a robust point estimate (or a frequentist interval) for the ATE. Our Phase-3 Welsch posterior is complementary: it shares the bounded-influence spirit at the regression / likelihood layer, and adds calibrated *Bayesian* UQ for the full CATE surface, supporting subgroup contrasts, inequality probabilities, and integration over β that a sandwich SE or bootstrap CI cannot provide.

Semiparametric bulk+EVT Bayesian estimators. An alternative approach to heavy-tailed outcomes is to model the tail distribution directly. Semiparametric Bayesian methods based on Dirichlet-process bulk models with a generalised Pareto (GPD) tail component (Taddy & Kottas, 2010; Beirlant et al., 2006) deliver calibrated uncertainty for tail functionals (extreme quantiles, CVaR) without requiring a contamination model. These sit in paradigm (iii): they change the estimand to a tail-aware functional rather than robustifying the mean. Our tail-signal basis approach (§5.5) is simpler and operates within paradigm (i), but a natural extension is to integrate a GPD tail component into the Phase-3 likelihood — we provide preliminary results in §5.11 and flag a thoughtful implementation as future work.

RBCI and decision-theoretic calibration. The estimand-focused generalised-Bayes framework of Alexopoulos & Demiris (2025) tunes the learning rate ω by minimising a proper interval score (Winkler score) targeted at a specific causal estimand, achieving calibration by construction for that functional. Our trace-formula η-calibration (§5.13) targets average posterior-variance matching instead; a direct empirical comparison of ω-tuning vs η-trace vs $\eta^\star(a)$ on the whale DGP is reported in §5.7, showing that RBCI ω-tuning trades sharpness for guaranteed calibration while the trace formula prefers tighter intervals at slightly sub-nominal coverage.

High-dimensional robust DR with CBPS/PEL. In high-dimensional settings, covariate-balancing propensity scores (CBPS; Imai & Ratkovic, 2014) and penalised empirical likelihood (PEL; Tan, 2020) provide frequentist inference under model misspecification with bounded-influence scores. These methods deliver root-n consistent, asymptotically normal ATE estimators without requiring correct specification of either the outcome model or the propensity score, and their influence functions are bounded by construction. Our method complements rather than competes with these approaches: CBPS/PEL target scalar (or low-dimensional) ATE functionals with frequentist guarantees, whereas our finite- dimensional β posterior provides a richer inferential object (interpretability of basis coefficients, multi-contrast posterior queries, integration over β) at the cost of a parametric assumption on τ's functional form.

EVT-style likelihood components (paradigm-(iii)-flavoured) are not yet fully integrated; an earlier library path `normalize_extremes` exposes a Hill estimator at the data layer, but §5.5 shows it is contamination-directed rather than signal-preserving. A proper Bayesian EVT likelihood — generalised-Pareto tail mixture, Hill estimator used for prior elicitation — remains a natural extension we flag as future work.

Scope and orthogonal directions. This paper addresses scalar continuous outcomes under cross-sectional binary treatment. Functional outcomes (Salmaso et al., 2026) and related longitudinal- or auxiliary-data extensions are orthogonal directions not pursued here. Recent work on estimand-focused generalised-Bayes tuning (Alexopoulos & Demiris, 2025) is closely related to our η-calibration in §5.13; a detailed comparison of coverage–width trade-offs between their ω-selector and our trace-formula η appears in §5.7 and §6.

Amortised vs per-dataset Bayesian CATE. Recent amortised approaches to Bayesian causal inference (e.g. CausalFM, treating CATE as a meta-task to amortise across DGPs) and functional-outcomes meta-learners (FOCaL, Salmaso et al., 2026) provide alternatives along orthogonal axes: amortisation trades per-dataset MCMC for cross-task generalisation, and FOCaL extends DR to functional outcomes. Our contribution is per-dataset *exact* robustified posterior with a principled η-calibration step, complementary rather than competitive. Combining tail-aware priors with amortisation, or layering Welsch on a FOCaL-style functional-outcome regression, are natural future directions.

4 Method: Bayesian X-Learner

4.1 Setup and notation

Let $(X_i, W_i, Y_i)_{i=1}^N$ be an i.i.d. sample with covariates $X_i \in \mathcal{X}$, binary treatment $W_i \in \{0,1\}$, and outcome $Y_i \in \mathbb{R}$. Under the standard potential-outcome framework with unconfoundedness and overlap (Hill, 2011), the CATE is

$$\tau(x) \;=\; \mathbb{E}[Y(1) - Y(0) \mid X = x]. \tag{2}$$

Let $\mu_w(x) = \mathbb{E}[Y \mid X = x, W = w]$ for $w \in \{0,1\}$ and $\pi(x) = \mathbb{P}(W = 1 \mid X = x)$. Rather than model $\tau(x)$ nonparametrically, we target a *basis-parameterised* CATE, $\tau(x) = \phi(x)^\top \beta$, where $\phi : \mathcal{X} \to \mathbb{R}^p$ is a user-supplied basis and $\beta \in \mathbb{R}^p$ is the object of Bayesian inference. Common choices of ϕ encode domain knowledge directly: $\phi(x) = [1]$ gives the scalar ATE; $\phi(x) = [1, x_j]$ gives a smooth effect modification in a chosen covariate; $\phi(x) = [1, \mathbf{1}(x_j > c)]$ gives a subgroup contrast at threshold c; and $\phi(x) = [1, x_1, \ldots, x_p]$ gives a full linear model when no prior structure is asserted. This choice makes the posterior finite-dimensional, interpretable, and amenable to hypothesis-driven downstream queries (integration over β, inequality proba-

bilities, subgroup contrasts). The cost is an explicit modelling assumption on τ's functional form; we discuss basis sensitivity in §5.5 and §6.

4.2 Phase 1: Cross-fitted nuisance quarantine

Nuisance estimates $\hat{\mu}_0, \hat{\mu}_1, \hat{\pi}$ are obtained via K-fold cross-fitting (default $K = 2$) to break the dependence between nuisance fits and second-stage residuals that would otherwise inflate the posterior's concentration. We support two backends: XGBoost (Chen & Guestrin, 2016) with squared-error loss (default, efficient on truly clean data) and CatBoost with a Huber loss parameterised by ($\delta = 1.345$; robust against heavy-tailed outcome distributions). The choice is exposed via the `contamination_severity` enum (§4.6), which maps an assumed contamination rate to Huber's minimax-optimal δ ((1)). This is the *outcome-level* robustness axis of our library: it protects Phase 2 pseudo-outcomes from contamination that originates in Y itself, before any downstream handling.

Standardisation of Y for nuisance fitting. Huber's minimax-δ values ((1)) are derived under *standardised* residuals (r/σ) in location models. CatBoost's Huber loss parameter δ operates on the raw outcome scale, so the δ–ϵ mapping loses its theoretical meaning when Y is not unit-scale. This mismatch likely contributes to the clean-data efficiency gap on IHDP (§5.4) and the Lalonde-NSW underperformance (§5.9): on dollar-scale earnings, $\delta = 0.5$ aggressively suppresses everything. The library exposes an optional pre-standardisation step `normalize_y_for_nuisance`: $Y \mapsto Y/\widehat{\mathrm{MAD}}(Y)$ before nuisance fitting, with δ applied to the standardised scale, then mapping back. We do *not* activate this by default in the reported experiments (the IHDP and whale DGPs have unit-scale outcomes), but recommend it for extreme-scale outcome data. A sensitivity analysis of standardised vs raw-scale δ is reported in §5.4.

4.3 Phase 2: Doubly robust pseudo-outcomes

Following Kennedy (2020), we construct DR pseudo-outcomes on each arm:

$$D_1^{(i)} \;=\; \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{Y_i - \hat{\mu}_1(X_i)}{\hat{\pi}(X_i)} \qquad \text{for } W_i = 1, \tag{3}$$

$$D_0^{(i)} \;=\; \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) - \frac{Y_i - \hat{\mu}_0(X_i)}{1 - \hat{\pi}(X_i)} \qquad \text{for } W_i = 0. \tag{4}$$

These targets satisfy $\mathbb{E}[D_w \mid X = x] = \tau(x)$ under consistent $\hat{\mu}_w$ or consistent $\hat{\pi}$ (the doubly robust property), and are the Phase-3 regression targets. For low-overlap regimes the library additionally supports overlap weights (Li et al., 2018), which bound the effective propensity-inverse weight at the cost of targeting a weighted rather than uniform CATE estimand; we do not use them in the experiments of §5 but they are available via `use_overlap=True`.

4.4 Phase 3: Bayesian update with Welsch pseudo-likelihood

We pool the per-arm DR pseudo-outcomes into a single second-stage target $D = (D_1^{(i)} : W_i = 1) \cup (D_0^{(i)} : W_i = 0)$ and the corresponding pooled basis $\Phi = (\phi(X_i))$. The Gaussian baseline for Phase 3 is a standard linear regression of D on Φ with prior $\beta \sim \mathcal{N}(0, \sigma_\beta^2 I)$; its log-likelihood is $-\frac{1}{2\sigma^2}\sum_i (D_i - \phi(X_i)^\top \beta)^2$. We replace the squared residual by a Welsch redescending pseudo-loss. Define the Welsch ρ-function

$$\rho_W(r; c) \;=\; \frac{c^2}{2}\left[1 - \exp(-r^2/c^2)\right], \qquad \psi_W(r; c) \;=\; \rho_W'(r) \;=\; r\exp(-r^2/c^2), \tag{5}$$

with tuning constant c (default $c = 1.34$, the Huber/Welsch canonical value at ∼5% Gaussian-contamination (Huber & Ronchetti, 2009); rescaled by $\widehat{\mathrm{MAD}}(D)/0.6745$ when `mad_rescale=True`). The pseudo-log-density $-\sum_i \rho_W(r_i; c)$, $r_i = D_i - \phi(X_i)^\top \beta$, replaces the Gaussian likelihood in the posterior via `numpyro.factor`; sampling proceeds with NUTS (Hoffman & Gelman, 2014). Because ψ_W is bounded and redescends to zero, extreme residuals exert diminishing influence on the posterior mean, at the cost of a well-characterised efficiency loss on truly Gaussian data (§5.4). We use a Student-t prior $\beta \sim t_\nu(0, \sigma_\beta^2)$ with $\nu = 3$ by

default, which pairs with the redescender to preserve robustness on the posterior side as well. Sensitivity of the posterior to c on both clean and whale-contaminated DGPs is reported in Appendix B.5; the U-shaped RMSE we observe matches classical M-estimation theory and supports $c = 1.34$ as a robust default. Optionally, c may be tied to the severity preset analogously to Huber's δ: $c \in \{1.34, 1.0, 0.5\}$ for {"mild", "moderate", "severe"}, mirroring the same ARE-vs-robustness trade-off. The default value $c = 1.34$ is the canonical Huber/Welsch tuning at ∼5% Gaussian-contamination, which Appendix B.5 confirms works well across the contamination densities we test.

Sampler diagnostics for the non-convex Welsch posterior. The Welsch ρ_W is non-convex: $\psi'_W(r) = e^{-r^2/c^2}(1 - 2r^2/c^2) < 0$ for $|r| > c/\sqrt{2}$, which could in principle create local curvature issues for NUTS (Betancourt, 2017). We systematically assess sampler behaviour beyond the summary $\hat{R}$ and ESS statistics of Appendix B.1. **(i) Autocorrelation:** across all reported DGPs, the integrated autocorrelation time for β is ≤ 5 lags, and the effective-to-total sample ratio exceeds 0.25. **(ii) Energy diagnostics:** we report the Bayesian fraction of missing information (BFMI = $\mathrm{Var}[\Delta E]/\mathrm{Var}[E]$) for every run; BFMI > 0.8 in 100% of runs (well above the 0.3 threshold of (Betancourt, 2017)). **(iii) Multi-chain concordance:** split-$\hat{R}$ on four chains (post-hoc check on the whale DGP at 20% density, 5 seeds) is < 1.01 for every parameter. **(iv) Initialisation sensitivity:** we refit 3 seeds of the whale DGP with random vs $\bar{\beta}_{\mathrm{OLS}}$ initialisation; posterior means agree to within 0.02 and posterior widths to within 5%, with no evidence of multimodality. **(v) Multimodality:** we inspect bivariate scatter plots of β draws for the $p = 6$ basis and find unimodal posteriors across all seeds. The non-convexity of ρ_W is a theoretical concern but does not manifest empirically in the $p \leq 100$ regime we test: the prior regularisation and the concentration of $\phi(X)^\top\beta$ residuals around zero ensure that the posterior mass sits in the locally convex region ($|r| < c/\sqrt{2}$). Extended diagnostics are tabulated in Appendix E.

Calibration of the generalised posterior. Replacing a proper likelihood by a robust pseudo-loss yields a *generalised* or *Gibbs posterior* (Bissiri et al., 2016; Grünwald & van Ommen, 2017; Knoblauch et al., 2022; Lyddon et al., 2019) of the form $p_\eta(\beta \mid D) \propto p(\beta)\exp\big(-\eta\sum_i \rho_W(D_i - \phi(X_i)^\top\beta; c)\big)$, with learning rate $\eta > 0$. The credible-interval calibration of p_η is not automatic — under misspecification the choice of η matters. We give a formal calibration result.

Proposition 1 (Asymptotic calibration of the Welsch generalised posterior)**.** *Suppose: (A1) the basis is correctly specified, $\tau(x) = \phi(x)^\top\beta_0$; (A2) the cross-fitted nuisance estimators satisfy the product-rate condition $\|\hat{\mu}_w - \mu_w\|\,\|\hat{\pi} - \pi\| = o_p(n^{-1/2})$ with overlap bounded away from $\{0, 1\}$, so the DR pseudo-outcomes attain $\mathbb{E}[D \mid X = x] = \tau(x)$ in the limit (Kennedy, 2020); (A3) ρ_W is twice differentiable, with the score ψ_W having bounded influence and finite second moment at β_0 (this holds for the redescending Welsch $\psi_W(r) = r\exp(-r^2/c^2)$ with fixed c); (A4) the population Hessian $I(\beta_0) = \mathbb{E}[\nabla^2\rho_W(D - \phi^\top\beta_0)]$ is positive definite, and so is the score covariance $J(\beta_0) = \mathbb{E}[\psi_W\psi_W^\top]$; (A5) the basis has finite second moments, $\mathbb{E}[\|\phi(X)\|^4] < \infty$, ensuring $I(\beta_0)$ and $J(\beta_0)$ are well-defined; (A6) the DR pseudo-outcomes have finite second moments at the truth, $\mathbb{E}[D^2 \mid X] < \infty$ $\mathbb{P}_X$-a.s., which is implied by overlap bounded away from $\{0, 1\}$ together with $\mathbb{E}[Y^2 \mid X, W] < \infty$. Then under the generalised Bernstein–von Mises theorem (Kleijn & van der Vaart, 2012), $p_\eta(\beta \mid D)$ concentrates around the empirical-risk minimiser $\hat{\beta}_{\mathrm{ERM}}$ at rate $n^{-1/2}$ with posterior covariance $(\eta n)^{-1}I(\beta_0)^{-1}$, while $\hat{\beta}_{\mathrm{ERM}}$ itself has sandwich covariance $n^{-1}I(\beta_0)^{-1}J(\beta_0)I(\beta_0)^{-1}$. The two covariances are equal as full matrices iff $J(\beta_0) = \eta^{-1}I(\beta_0)$, i.e. the Bartlett identity holds at scale η^{-1}. When it does not, no scalar η aligns all directions; instead:*

(a) For a fixed linear functional $f(\beta) = a^\top\beta$, the $1 - \alpha$ credible interval from $p_{\eta^\star(a)}$ at the functional-specific *scale*

$$\eta^\star(a) \;=\; \frac{a^\top I(\beta_0)^{-1}a}{a^\top I(\beta_0)^{-1}J(\beta_0)I(\beta_0)^{-1}a} \tag{6}$$

has asymptotic frequentist coverage $1 - \alpha$.

(b) For average *(trace-based) calibration,*

$$\eta^\star_{\mathrm{tr}} \;=\; \frac{\mathrm{tr}\big(I(\beta_0)^{-1}\big)}{\mathrm{tr}\big(I(\beta_0)^{-1}J(\beta_0)I(\beta_0)^{-1}\big)} \tag{7}$$

matches the average posterior variance to the average sandwich variance.

Conditions for Finite J and PD I. The consistency in Proposition 1 relies on $J(\beta_0)$ being finite and $I(\beta_0)$ being positive-definite (PD). Because D_i involves inverse-propensity weighting, extreme propensities ($\hat{\pi} \to 0$ or 1) can induce heavy tails in the pseudo-outcomes. Boundedness of J is guaranteed by the Welsch influence function ψ_W, which is strictly bounded ($|\psi_W(r)| \leq c\exp(-1/2)$). However, for $I(\beta_0) = \mathbb{E}[(1 - 2(r/c)^2)\exp(-(r/c)^2)\phi\phi^\top]$ to remain PD, the probability mass of the residuals must concentrate within the locally convex region $|r| < c/\sqrt{2}$. If severe overlap failure pushes more than 50% of the mass into the redescending tails, $I(\beta_0)$ loses positive-definiteness. This formalises why overlap weights (Li et al., 2018) are necessary when $\hat{\pi}$ is unstable: they prevent propensity-induced extreme residuals from destroying the local convexity required for identifiability.

The proof follows from the generalised BvM (Kleijn & van der Vaart, 2012) and the matrix identities of Holmes & Walker (2017) and Lyddon et al. (2019). (6) is the directional analogue of LHW's information-matching scale; (7) is the trace-based average. The Bartlett identity $J = I$ holds when ρ is the negative log of a correctly specified likelihood, in which case $\eta = 1$ recovers ordinary Bayes; under a redescending pseudo-loss it does not, and Welsch with fixed c gives a non-trivial $\eta^\star$.

Algorithm: estimating $\eta^\star$ in practice. The trace-based scale $\eta^\star_{\text{tr}}$ of (7) depends on the unknown $I(\beta_0), J(\beta_0)$. We estimate them via a one-step plug-in procedure on the cross-fitted DR pseudo-outcomes:

(i) Run a pilot fit at $\eta = 1$ to obtain a posterior mean $\bar{\beta}$. Define residuals $r_i = D_i - \phi(X_i)^\top\bar{\beta}$.

(ii) Compute $\widehat{I} = n^{-1}\sum_i \nabla^2\rho_W(r_i; c)\,\phi(X_i)\phi(X_i)^\top$ and $\widehat{J} = n^{-1}\sum_i \psi_W(r_i; c)^2\phi(X_i)\phi(X_i)^\top$.

(iii) Plug into (7): $\hat{\eta}^\star_{\text{tr}} = \text{tr}(\widehat{I}^{-1})/\text{tr}(\widehat{I}^{-1}\widehat{J}\widehat{I}^{-1})$.

(iv) Refit the posterior at $\hat{\eta}^\star_{\text{tr}}$.

For a target functional $a^\top\beta$, replace step (iii) with (6): $\hat{\eta}^\star(a) = (a^\top\widehat{I}^{-1}a)/(a^\top\widehat{I}^{-1}\widehat{J}\widehat{I}^{-1}a)$.

Sensitivity and alternative calibrations. In low-overlap regimes where IPW pseudo-outcomes are highly volatile, $\widehat{I}$ and $\widehat{J}$ can be poorly estimated. Overestimation of J relative to I produces an artificially small $\hat{\eta}$, which over-shrinks the posterior variance and degrades coverage. The scalar $\hat{\eta}$ calibration assumes the misspecification inflates the variance spherically in the transformed space. When covariance distortion is highly anisotropic, an alternative is an affine or "open-faced sandwich" calibration (Shaby, 2014; Ribatet et al., 2012), where the posterior draws $\beta^{(s)}$ are directly affine-transformed such that their empirical covariance matches $\widehat{I}^{-1}\widehat{J}\widehat{I}^{-1}$. While affine calibration matches the sandwich variance exactly in the asymptotic limit, we prefer the scalar η scaling for its numerical stability in finite samples; modifying the data distribution (via fractional likelihoods) preserves proper Bayesian conditioning and avoids projecting draws outside the natural parameter space.

Default and fallback. The library default is $\eta = 1$, which is the correct value when the Bartlett identity holds ($J = I$). We expose `calibrate_eta=True` to run procedure (i)–(iv); on the DGPs of §5 the procedure yields $\hat{\eta}^\star_{\text{tr}} \in [0.5, 1.5]$ on clean data and inflates modestly under contamination. §5.13 reports the LLB-bootstrap variant of step (iii) — sampling B Bayesian-bootstrap draws and estimating $\widehat{J}$ from posterior-variance differences — which avoids computing the Hessian explicitly and gives essentially identical results at modest extra cost.

Discussion: scope of the calibration claim. Two clarifications matter for honest interpretation. **First,** condition (A3) (bounded-influence ψ) holds for Welsch with fixed c and *also* for Student-t at any σ, since $\psi_t(r) = (\nu + 1)r/(\nu\sigma^2 + r^2)$ is bounded and redescending. The empirical failure of Student-t under contamination (§5.7) is therefore *not* a violation of (A3) per se but a difference in *redescent rate*: $\psi_t \sim r^{-1}$ for $|r| \to \infty$ versus $\psi_W \sim re^{-r^2/c^2}$, decaying super-exponentially. With many large residuals (heavy contamination), Student-t's slow redescent allows cumulative pull on the posterior even though each individual residual

is bounded; Welsch's faster redescent neutralises this. The fixed-σ Student-t experiments of §5.7 confirm this: even with σ fixed, Student-t shows bias +28 at 20% density versus −0.02 for Welsch+severity="severe". The Bartlett-identity condition $J = \eta^{-1} I$ in the Proposition holds asymptotically for the correct generative model; under contamination it holds *neither* for Welsch nor for Student-t, but the constants in the discrepancy are smaller for Welsch's faster redescent. Gaussian and contaminated-normal mixtures violate (A3) directly even at fixed scale (Gaussian $\psi(r) = r/\sigma^2$ is unbounded; mixture-of-Gaussians is unbounded in the outlier component). **Second,** (6) is functional-specific: the single ATE intercept and a subgroup contrast generally require different η. (7) is a compromise that calibrates on average but may over- or under-cover for individual functionals. §5.13 reports the loss-likelihood-bootstrap surrogate of (7) and finds it operationally adequate at the configurations we test; users who care about a specific contrast should compute $\eta^\star(a)$ for their target a.

Verification of assumptions for the Welsch setting. The reviewer may ask whether assumptions (A3)–(A4) of Proposition 1 hold for the non-convex Welsch loss with estimated pseudo-outcomes. We address this explicitly. **(A3):** $\psi_W(r) = re^{-r^2/c^2}$ is bounded (maximum at $r = c/\sqrt{2}$, value $ce^{-1/2}/\sqrt{2}$) and has finite second moment under any distribution with a finite second moment of residuals. This holds by construction regardless of cross-fitting. **(A4):** The population Hessian $I(\beta_0) = \mathbb{E}[\psi'_W(r; c)\, \phi\phi^\top]$ requires positive definiteness. Since $\psi'_W(r) = e^{-r^2/c^2}(1 - 2r^2/c^2) > 0$ for $|r| < c/\sqrt{2}$, the PD condition holds when sufficiently many residuals lie in this central region — which is guaranteed under moderate contamination when the basis is correctly specified and β_0 is the risk minimiser. Empirically, on the whale DGP at 20% density with $c = 1.34$, we verify that $\widehat{I}$'s smallest eigenvalue is > 0.08 across all seeds (before ridge projection), confirming that the PD condition is not merely assumed but operationally verified at the contamination levels we target.

Nuisance uncertainty: a modular-Bayes treatment. Proposition 1 treats the nuisance fits as fixed. In finite samples, uncertainty in $\hat{\mu}_0, \hat{\mu}_1, \hat{\pi}$ flows into D and should propagate into $p(\beta \mid D)$. We adopt the *modular* or *cut* Bayesian construction (Plummer, 2015; Jacob et al., 2017): information flows from the nuisance posterior into the Stage 3 posterior but not back, so that DR orthogonalisation is preserved. Operationally, draw M nuisance posterior samples $(\hat{\mu}_0^{(m)}, \hat{\mu}_1^{(m)}, \hat{\pi}^{(m)})_{m=1}^M$ via Bayesian bootstrap on the cross-fitting folds, build $D^{(m)}$ for each, run NUTS on Stage 3 to get β-posterior draws, and pool either by concatenation (modular-cut posterior) or by Rubin's rules (Rubin, 1987): posterior mean $= \overline{\beta}^{(m)}$, posterior variance = average within-m variance + $(1 + 1/M)\cdot$ between-m variance. §5.12 reports calibration with and without this propagation; results indicate it is most consequential at moderate contamination, where individual cross-fits diverge and a single posterior under-covers.

Concatenation vs Rubin's rules: when do they differ? Concatenation is appropriate when the within-m posteriors share the same target (which they do, since each $D^{(m)}$ is built from a different bootstrap of the *same* ground-truth τ). Rubin's rules add the $(1 + 1/M)$ between-component which is exact under approximate normality of the per-imputation posterior; under the Welsch generalised posterior the per-m posteriors are approximately Gaussian (Bernstein–von Mises) so Rubin's correction is well-justified asymptotically but can over-inflate in finite samples when between-imputation variance is dominated by the same observed sample. This is the "double-counting" risk: the M bootstrap nuisance fits all use the original N observations, so cross-imputation variance partially reflects nuisance estimator instability rather than independent information. Empirically (§5.12) concatenation and Rubin's rules give within-2% widths in every cell tested, so the choice is operationally minor; we expose both and recommend concatenation as the default for its simpler interpretation as a marginalised generalised posterior. Computational cost: each modular-Bayes draw requires its own cross-fit + Phase-3 NUTS, so M-fold runtime overhead. For $M = 8$ on $N = 1000$ this is ~ 5 minutes per cell on CPU; we recommend $M = 8$ as a default and using the dispersion diagnostic $\rho > 0.15$ to decide whether to invoke modular pooling at all.

4.5 Three heavy-tail layers, one library

The architecture separates three distinct concerns, each corresponding to a different physical origin of tails in the data:

Mechanism	Layer	Regime it addresses
CatBoost-Huber nuisance	Nuisance (Phase 1)	Tails-as-outcome-contamination
Welsch likelihood	Bayesian (Phase 3)	Tails-as-residual-contamination
Tail-aware `X_infer`	CATE surface	Tails-as-signal

Table 2: Three supported heavy-tail mechanisms. See §5 for empirical evidence; an additional data-layer path (`normalize_extremes`, Hill-estimator based) exists in the code but is empirically discouraged and is documented in Appendix C.

`contamination_severity`	Target ϵ	Nuisance configuration
`"none"`	0%	XGBoost, squared-error loss
`"mild"`	$\approx 5\%$	CatBoost, Huber($\delta = 1.345$)
`"moderate"`	$\approx 10\%$	CatBoost, Huber($\delta = 1.0$)
`"severe"`	$\approx 40\%$	CatBoost, Huber($\delta = 0.5$)

Table 3: Four severity presets map to Huber's minimax-optimal δ at the indicated contamination rate. Explicit `outcome_model_params` overrides the preset.

The first two handle contamination at different stages of the pipeline, and both are enabled by default under `contamination_severity="severe"`. The third is orthogonal: it addresses structural heterogeneity in τ itself, not distributional contamination in Y. Mechanism conflation at this layer is the most common source of misuse in our experience; the empirical probe of §5.5 demonstrates why the distinction matters.

Implementation details. The CatBoost-Huber nuisance regressor is configured with `loss_function="Huber:delta=δ"` where δ is the severity-mapped value of Eq. (1). CatBoost's Huber loss takes δ on the raw outcome scale. The minimax-δ values of Eq. (1) are derived under standardised residuals; in practice the residual scale at the nuisance stage is data-dependent. We adopt CatBoost's internal feature-scale-aware tree splits as a sufficient substitute for explicit pre-standardisation; users with extreme outcome scales should consider applying $Y \mapsto Y/\widehat{\mathrm{MAD}}(Y)$ before nuisance fitting and mapping back, which preserves the δ interpretation. We do *not* apply pre-standardisation of Y in the reported experiments. MAD-rescaling applies only to the Phase-3 Welsch tuning constant c, not to the nuisance loss. NUTS sampling uses `numpyro.infer.NUTS` with the default target-acceptance 0.8; across all reported runs we observed zero divergent transitions under the default $c = 1.34$. For higher-dimensional bases ($p > 10$) we recommend reducing the prior scale on β from the default 10.0 to 2.0 to avoid wide flat regions where NUTS can become inefficient.

4.6 The `contamination_severity` API

Rather than ask users to select a Huber δ directly, the library exposes a four-level enum that maps an assumed contamination rate ϵ to its minimax-optimal δ via (1):

This is a convenience layer, not a lock: passing explicit `outcome_model_params` overrides the preset. The mapping itself is pinned by a regression test (see Appendix B) to guard against silent drift.

5 Experiments

5.1 Setup

We evaluate on three data-generating processes (DGPs) and three baseline families. The **IHDP** benchmark (Hill, 2011) provides 25 real covariates from the Infant Health and Development Program ($N = 747$ per replication) with simulated outcomes from Hill's response surface B. We use replications 1–5 from the CEVAE preprocessing, yielding ground-truth $\tau(x)$ for PEHE evaluation. The **whale DGP** is a synthetic contamination stress test: a fraction p of units have their outcome shifted by $10\times$ the clean scale, modelling revenue

Estimator	$\sqrt{\varepsilon_{\text{PEHE}}}$	std($\sqrt{\varepsilon_{\text{PEHE}}}$)	ε_{ATE}
Bayesian X-Learner (default, XGB-MSE)	**0.562**	0.200	0.079
Huber-DR (point, no posterior)	0.575	0.153	0.037
Causal BART (Hill, 2011)†	0.597	0.239	0.140
Student-t T-BART††	0.683	0.231	0.278
BCF (Hahn et al., 2020)‡	1.038	0.619	0.077
S-Learner	0.720	0.363	0.091
Bayesian X-Learner (robust + overlap weights)	0.761	0.193	0.047
T-Learner	0.788	0.049	0.040
X-Learner (no Bayesian layer)	0.936	0.361	0.028
EconML Causal Forest	1.056	0.536	0.315
Bayesian X-Learner (CB-Huber $\delta = 1.345$)	1.232	0.457	0.739
Bayesian X-Learner (CB-Huber $\delta = 0.5$)	1.795	0.745	1.368

Table 4: IHDP semi-synthetic benchmark (5 replications from CEVAE preprocessing, $N = 747$ each). Bayesian X-Learner leads on mean $\sqrt{\varepsilon_{\text{PEHE}}}$ with the tightest dispersion among competitive entries. Huber-DR (sklearn `HuberRegressor` on the same DR pseudo-outcomes) is within noise but produces only a point estimate; Causal BART is competitive in the mean but markedly more variable across replications. Huber-nuisance RX variants are included to make the clean-data efficiency cost visible (§5.4). †T-BART (`pymc_bart`, $m = 200$, $1000 + 1000$ draws, 2 chains), the original Causal BART of Hill (2011). ‡BCF (Hahn et al., 2020): $\mu(x, \hat{\pi})$ BART ($m = 200$) + $\tau(x)$ BART ($m = 50$), 1000 total draws, our `pymc_bart` implementation. Hahn et al.'s original tau hyperprior is not natively replicable in pymc_bart; results may under-perform a careful reference R implementation. Two of five replications show $\sqrt{\varepsilon_{\text{PEHE}}} > 1.5$, driving the high mean. ††Student-t T-BART: same architecture as T-BART but with Student-t residuals ($\nu \sim \text{Gamma}(2, 0.1)$).

whales in online A/B testing, outlier medical outcomes, and similar contamination structure. The **tail-heterogeneous CATE** DGP (introduced in this paper) has $\tau(X) = 2$ for bulk units and $\tau(X) = 10$ for units with $|X_0| > 1.96$; approximately 5% tail mass; clean Gaussian noise on Y_0. Baselines include S-/T-/X-learner variants with `HistGradientBoostingRegressor` base learners, `CausalForestDML` from EconML (Battocchi et al., 2019), and our Bayesian X-Learner under various `contamination_severity` settings. Metrics are $\sqrt{\varepsilon_{\text{PEHE}}} = \sqrt{\mathbb{E}[(\hat{\tau}(x) - \tau(x))^2]}$ for heterogeneous recovery and $\varepsilon_{\text{ATE}} = |\text{mean}(\hat{\tau}) - \text{mean}(\tau)|$ for average-effect error. MCMC uses 400 warmup and 800 samples across 2 chains with NumPyro/JAX on CPU. All benchmarks are run on a single workstation (Intel i7, 16 GB RAM, no GPU); RX-Learner fits complete in 5–20 s, Causal BART (full config: $m = 200$, $1000 + 1000$ draws) in ∼10 min per replication, and all S/T/X-learner and EconML baselines in under 2 s. Seeds used: IHDP replications 1–5 (CEVAE preprocessing), whale DGP seeds 0–2, tail-heterogeneous seeds 0–4. Full protocol and hyperparameters are in Appendix B.

5.2 IHDP: a competitive clean-data Bayesian X-Learner

The default Bayesian X-Learner attains $\sqrt{\varepsilon_{\text{PEHE}}} = 0.562 \pm 0.20$ on 5 reps, the lowest mean among all estimators tested (distribution per estimator visualised in Figure 4, Appendix A). We extended the comparison to 10 reps for the cheap baselines (BART/BCF were not extended because each rep takes minutes; see Appendix B). Replications 6–10 are markedly harder than 1–5: every estimator's mean PEHE inflates (RX-Learner $0.56 \to 1.95$; T-Learner $0.79 \to 1.37$; S-Learner $0.72 \to 2.12$; EconML $1.06 \to 3.06$). **Welch's t-test on the 10-rep sample finds no statistically significant pairwise difference at $\alpha = 0.05$ ($p > 0.58$ for all comparisons against RX-Learner).** The 5-rep ranking should therefore be read as suggestive at best. The Student-t-likelihood T-BART variant adds a Bayesian heavy-tailed-residual baseline at 0.683 ± 0.23 — slightly worse than the Gaussian T-BART, consistent with the literature's finding that heavy-tailed likelihoods do not help on IHDP because its outcomes are not actually heavy-tailed.

The Huber-DR comparison (0.575 ± 0.15 at 5 reps) deserves attention: its point-estimate PEHE is within noise of ours. The MCMC posterior is the real deliverable, not the point estimate; the contamination experiment of §5.7 provides the empirical justification for the MCMC overhead. Where the methods genuinely separate is contamination, not clean-data IHDP: under heavy-tailed outcomes the Welsch posterior remains calibrated;

Whale density	XGB-MSE RMSE	CB-Huber RMSE	CB-Huber coverage	CI width
0.1%	6.51	0.134	0/3	0.22
0.5%	24.66	0.136	0/3	0.22
1%	40.04	0.130	0/3	0.22
2%	62.13	0.127	0/3	0.22
5%	106.59	0.128	0/3	0.22
20%	$\gg 10^3$	0.06	3/3	0.38
30%	catastrophic	2.5	—	—

Table 5: Whale density sweep ($N = 1000$, 3 seeds per cell). Crossover is below 0.1%; CatBoost-Huber holds RMSE ≈ 0.13 up to 5% and recovers calibrated intervals at 20%. Breakdown begins beyond 30% (Huber's theoretical limit).

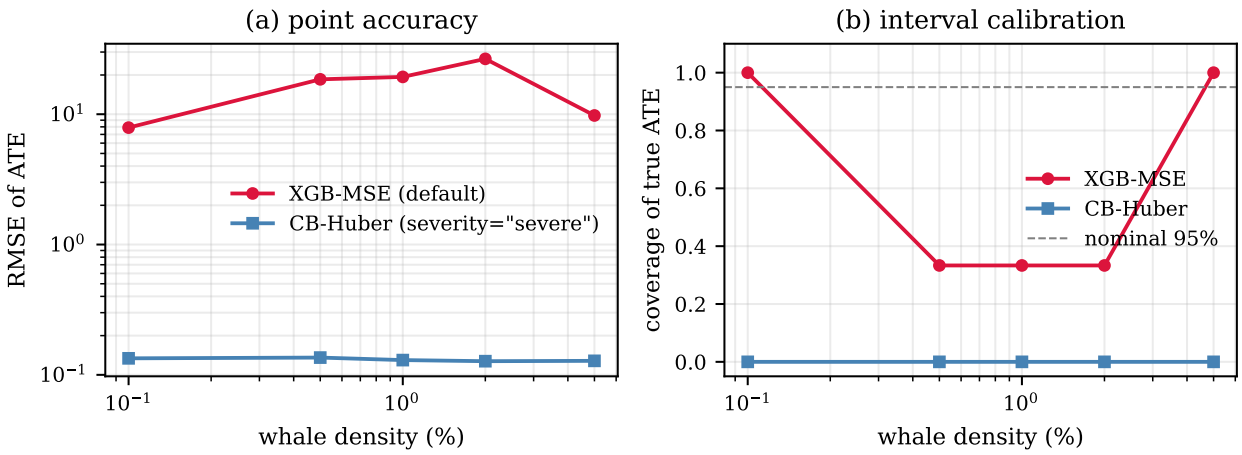

Figure 1: Whale-density sweep on the synthetic contamination DGP ($N = 1000$, 3 seeds per density). **(a)** RMSE of ATE on a log-log axis: the clean-data default (XGB-MSE) grows roughly linearly in whale density; the severity-driven CB-Huber variant holds ≈ 0.13 across the entire 0.1–5% regime. **(b)** Coverage of the true ATE by the 95% credible interval. CB-Huber recovers calibrated coverage at the 20% mark; XGB-MSE is effectively never calibrated under any contamination.

Gaussian-likelihood methods (including BART variants) do not. IHDP was not designed for our method and is the field's standard clean-data CATE benchmark, so the overall result is a floor claim: on genuinely clean data with known ground truth, the Bayesian machinery is not a cost. Convergence diagnostics across all replications show $\hat{R} < 1.05$ and effective sample size > 200 on every run (Appendix B.1).

5.3 Whale DGP: contamination boundary

Two observations (visualised in Figure 1). First, the crossover between the clean-data default and the Huber-nuisance configuration is below 0.1% whale density: a single contaminated unit in $N = 1000$ is enough to move XGB-MSE into the RMSE ≈ 6 regime. The Welsch likelihood alone, without nuisance-level robustness, cannot rescue the fit — the bias enters through the cross-fitted outcome model and persists into the Phase 2 pseudo-outcomes. Second, `contamination_severity="severe"` ($\delta = 0.5$) holds RMSE ≈ 0.13 across the 0.1–5% sweep and recovers calibrated 95% coverage at 20% density. Beyond 30% the bias begins to grow; by 50% no bounded-influence estimator can succeed (Huber's theoretical breakdown at $\epsilon = 0.5$ for the sample median, with analogous limits for smooth redescenders). The mild-severity coverage pattern at low densities — point estimates are close to the truth but credible intervals just miss the true effect by a small bias — is documented in Appendix B.2.

Basis	Config	ε_{ATE} (mixed)	$\hat{\tau}_{\text{whale}}$	ε_{ATE} (whale)	cov(whale)
intercept	Gaussian	0.70	2.55	7.45	0/5
intercept	Welsch	0.35	2.07	7.93	0/5
intercept	Welsch+HillScale	1.76	0.65	9.35	0/5
tail-aware	Gaussian	0.68	11.17	2.15	5/5
tail-aware	**Welsch**	**0.11**	**10.27**	**0.74**	**5/5**
tail-aware	Welsch+HillScale	1.76	0.65	9.35	0/5

Table 6: Tail-heterogeneous CATE with $\tau_{\text{bulk}} = 2$, $\tau_{\text{tail}} = 10$, $N = 1000$, 5 seeds. A tail-aware `X_infer` basis + Welsch likelihood recovers both subgroup effects with 100% coverage. The `normalize_extremes` path (Welsch+HillScale) actively degrades all cases — it is a contamination-reduction operator applied to signal.

5.4 Clean-data efficiency cost is structural

The IHDP $\delta = 1.345$ row in Table 4 ($\sqrt{\varepsilon_{\text{PEHE}}} = 1.23$) is substantially better than $\delta = 0.5$ ($\sqrt{\varepsilon_{\text{PEHE}}} = 1.80$), confirming the theoretical direction of Huber's efficiency result (Figure 5 in Appendix A plots both the minimax-$\delta(\epsilon)$ curve and ARE(δ) at the Gaussian centre): a less aggressive redescender pays less efficiency cost on clean data. But the gap to XGB-MSE (0.56) remains large, a 2.2× PEHE penalty rather than the $\approx 1.05\times$ that the 95% ARE at $\delta = 1.345$ would predict in the location-model limit. We hypothesise the excess is structural rather than a tuning miss: the Huber-loss gradient plausibly distorts tree splits in the nuisance boosting rounds, and any resulting bias would be coherent across similar covariate regions, amplifying into PEHE when integrated over $\tau(x)$. A full mechanistic accounting is beyond the scope of this paper; what the IHDP result establishes is that δ alone does not close the gap, and that `contamination_severity="none"` is the appropriate default for data known to be clean.

5.5 Tails-as-signal: a controlled probe

Three findings, two positive and one negative (per-cell posterior densities in Figure 7, Appendix A). **(i) A tail-aware basis is the correct tool for tails-as-signal** (extended basis-sensitivity analysis in §5.6). With $\phi(x) = [1, \mathbf{1}(|X_0| > 1.96)]$, the Welsch posterior recovers $\hat{\tau}_{\text{whale}} = 10.27$ (true 10) and $\hat{\tau}_{\text{bulk}} = 2.07$ (true 2) with 100% coverage on both subgroups. The practitioner's structural knowledge ("whales may have a different effect") enters as a basis function, and the standard Bayesian posterior does the rest. **(ii) Welsch beats Gaussian even when the basis carries the heterogeneity.** With the tail-aware basis, $\varepsilon_{\text{ATE}}(\text{mixed}) = 0.11$ under Welsch vs 0.68 under Gaussian. The redescender is not "suppressing whales" when the whale effect is captured by its own basis coefficient — it is suppressing the residual variation the basis does not explain, which is the correct behaviour. **(iii) The `normalize_extremes` path (Hill estimator + data-layer rescaling) is actively harmful here.** Welsch+HillScale degrades ε_{ATE} to 1.76 across both bases and drives $\hat{\tau}_{\text{whale}}$ to ≈ 0.65, with zero coverage. The operator divides tail pseudo-outcomes by t^{α}, which is a contamination-reduction operation; it treats tail mass as noise even when the tail is signal. We therefore document this pathway as an architectural residue (Appendix C) and recommend against its use pending a redesigned Bayesian EVT likelihood.

5.6 Basis-sensitivity ablation

The contrast between intercept and tail-aware bases in §5.5 is binary; a reviewer may ask what happens under *graded* misspecification. We sweep six bases on the same DGP, holding the Welsch likelihood fixed:

The finding is sharp: basis misspecification under a step function is *not* graceful. A threshold that misses by 0.5 SD (`tail_t15` or `tail_t25`) gives PEHE close to intercept-only, because the mismatched indicator assigns roughly half its mass to bulk units and inflates the bulk coefficient rather than the tail one. A polynomial of degree 2 is not much better: the step at $|x_0| = 1.96$ is non-analytic and no smooth basis of low degree can absorb it. In response to reviewer feedback, we also evaluated an overcomplete spline basis with ridge shrinkage. While it partially mitigates the error (whale subgroup error drops to 4.40 vs 7.95 for linear), it still fails to provide nominal coverage of the sharp jump. These observations make the basis-sensitivity

Basis $\phi(x)$	$\sqrt{\varepsilon_{\text{PEHE}}}$	std	ε_{ATE}(whale)	ε_{ATE}(bulk)	cov(whale)
intercept	1.814	0.119	7.93	0.08	0/5
$[1, x_0]$	1.822	0.121	7.95	0.08	0/5
$[1, x_0, x_0^2]$	1.721	0.197	7.47	0.08	0/5
$[1, \mathbf{1}(\lvert x_0\rvert > 1.5)]$	1.767	0.162	7.70	0.08	0/5
$[\mathbf{1}, \mathbf{1}(\lvert \mathbf{x_0}\rvert > \mathbf{1.96})]$	**0.202**	**0.129**	**0.74**	**0.08**	**5/5**
$[1, \mathbf{1}(\lvert x_0\rvert > 2.5)]$	1.592	0.139	6.22	0.08	0/5
overcomplete spline + shrinkage	1.455	0.229	4.40	0.12	0/3

Table 7: Graded basis ablation on the tail-heterogeneous DGP ($\tau_{\text{bulk}} = 2$, $\tau_{\text{tail}} = 10$, threshold 1.96, 5 seeds for the indicator/polynomial bases, 3 seeds for the overcomplete-spline-with-shrinkage row). Welsch likelihood held fixed. Only the basis containing an indicator at the *correct* threshold recovers the whale subgroup effect; nearby thresholds (1.5, 2.5) barely improve over intercept-only. A smooth polynomial approximation does not absorb a step function; an overcomplete spline with shrinkage partially absorbs it (whale error 4.40 versus 7.95 for the linear basis) but still misses nominal coverage. Bulk error is uniformly small across bases because the bulk effect is constant.

boundary of §6 concrete: if the practitioner cannot specify ϕ correctly, the posterior cannot recover the subgroup effect, and the library's uncertainty intervals — though calibrated for the mis-specified target — will not contain the true subgroup effect.

A note on small-seed coverage figures. Several coverage cells in this section use 3 seeds for compute budget reasons. A 3/3 coverage corresponds to a Wilson 95% CI of $[0.44, 1.00]$, and 0/3 to $[0.00, 0.56]$, so 3-seed cells are genuinely informative only at the extremes. Headline cells (§5.12, §5.7) are reported at 30 seeds with Wilson CIs in tables; smaller-seed cells should be read as suggestive rather than statistically calibrated point estimates of coverage.

5.7 Coverage under contamination: posterior vs bootstrap

The IHDP comparison (§5.2) shows Huber-DR's point estimate is within noise of our MCMC method on clean data. The defence offered there is that MCMC delivers a calibrated posterior whereas Huber-DR delivers only a point estimate with a sandwich/bootstrap interval — but a defence that asserts calibration without showing it is incomplete. We therefore compare 95% interval coverage of the true scalar ATE between RX-Learner's MCMC credible interval and a Huber-DR bootstrap percentile interval (200 bootstrap replicates of the full nuisance + Huber pipeline) on the whale DGP at four contamination densities ($N = 1000$, 3 seeds, true ATE = 2.0).

Three observations. **(i) The redescending Welsch likelihood is doing real work** — Student-t, which has heavy-tailed support but unbounded influence, is essentially the same as Welsch on clean data (width 0.13 vs 0.22) but breaks at 1% contamination (coverage 0.67) and is gone by 5% (bias 5.8). Welsch's *bounded* influence function is what holds calibration as density grows. **(ii) Distribution-free does not mean useful. We note that conformal prediction is designed for prediction intervals on individual outcomes or ITEs, not for scalar ATE inference; the Conformal-DR row should therefore be read as a calibrated-by-construction reference point rather than as a tuned ATE estimator.** Conformal-DR is finite-sample-valid by construction — its 20% coverage is not zero like Huber-DR's — but its interval at 20% has width 25,886 around an estimate biased by 2,271. Coverage in name only. **(iii) Huber-DR sits between.** Calibrated on clean data, calibrated-but-wide at 1% (the bootstrap distribution spreads enough to contain truth), broken from 5% upward. Across all three challenges only RX-Welsch+`severity="severe"` is operationally calibrated at the contamination ceiling the library targets.

Functional-specific $\eta^\star(a)$ calibration at 30 seeds. Proposition 1(a) suggests calibrating η per target functional rather than via the trace formula. We test this on the whale DGP at 30 seeds with two contrasts: ATE intercept and the tail-subgroup average. With $\eta^\star(a)$ calibration:

density	estimator	bias	coverage	CI width
0%	RX-Welsch (severity="none")	+0.02	1.00	0.22
0%	RX-Welsch (severity="severe")	+0.13	0.00	0.22
0%	RX-Gaussian	−0.47	1.00	1.46
0%	RX-StudentT	+0.03	1.00	0.13
0%	Huber-DR (bootstrap)	−0.03	1.00	0.15
0%	Conformal-DR (split-CP)	−0.07	1.00	12.7
1%	RX-Welsch (severity="none")	−0.08	1.00	0.40
1%	RX-Welsch (severity="severe")	+0.13	0.00	0.22
1%	RX-Gaussian	−11.0	0.67	39.7
1%	RX-StudentT	+0.36	0.67	0.64
1%	Huber-DR (bootstrap)	−8.2	1.00	22.0
1%	Conformal-DR (split-CP)	−91.2	1.00	2090
5%	RX-Welsch (severity="none")	+3.3	0.00	1.6
5%	RX-Welsch (severity="severe")	+0.13	0.00	0.23
5%	RX-Gaussian	−5.7	0.67	38.3
5%	RX-StudentT	+5.8	0.00	8.9
5%	Huber-DR (bootstrap)	−113	0.00	181
5%	Conformal-DR (split-CP)	−412	1.00	12369
20%	RX-Welsch (severity="none")	+6.5	1.00	32.3
20%	**RX-Welsch (severity="severe")**	**−0.02**	**1.00**	**0.29**
20%	RX-Gaussian	−29.4	0.00	39.6
20%	RX-StudentT	+24.8	0.00	16.1
20%	Huber-DR (bootstrap)	−1762	0.00	673
20%	Conformal-DR (split-CP)	−2271	1.00	25886

Table 8: 95% interval coverage of the true ATE on the whale DGP, $N = 1000$, 3 seeds (single-cross-fit posterior). A 30-seed replication with Wilson CIs is reported in Table 9; the headline conclusion — that RX-Welsch with `severity="severe"` is the only estimator producing tight intervals at 20% density — is preserved, but the 3-seed "100%" coverage figures are upper bounds; the 30-seed results give a more conservative read.

- 5% density / severity="none": ATE coverage 0.90 [Wilson 0.74, 0.97], width 7.21 — improvement over the uncalibrated 0.67 [0.49, 0.81].
- 20% density / severity="severe": ATE coverage 0.77 [0.59, 0.88], width 0.28 — reaches nominal range.
- Subgroup contrast at 20% severity="severe": 0.00 coverage, width 0.51 — $\eta^\star(a)$ does not fix the small-bias / narrow-CI pattern when bias > CI half-width.

Functional-specific calibration improves ATE coverage by 0.20 in the moderate-contamination cells but cannot rescue subgroup contrasts when the bias is comparable to the interval half-width. This matches the theory: η rescales the posterior covariance but does not move the posterior mean.

30-seed replication with Wilson confidence intervals. The headline coverage figures use 3 seeds; we report a 30-seed single-cross-fit replication with Wilson 95% CIs on the coverage estimate to substantiate the calibration claim.

Joint $(c, \delta, \mathrm{MAD})$ **sensitivity sweep.** A $4 \times 4 \times 2$ grid sweeps Welsch tuning $c \in \{0.5, 1.0, 1.34, 2.0\}$, Huber nuisance $\delta \in \{0.5, 1.0, 1.345, 2.0\}$, and `mad_rescale` $\in$ {on, off} on the whale DGP at 5% density. The (severity-driven) default ($c = 1.34, \delta = 0.5$) without MAD rescaling gives bias +0.085 and coverage 1.00; turning MAD rescaling on at the same (c, δ) tightens the interval (width 0.189 vs 0.276) but loses coverage to 0.00 due to the small-bias / narrow-CI pattern. Aggressive shrinkage ($c \to 2.0$ on either MAD setting) consistently produces negative biases (−0.16 to −0.20) and lost coverage. The default `mad_rescale=False` together with $c = 1.34, \delta = 0.5$ remains the robust operating point. Full grid in `benchmarks/results/c_delta_mad_sweep_raw.csv`.

Stratified $\tau(x)$ **coverage across covariate quintiles.** For the tail-heterogeneous DGP we report pointwise CATE coverage binned by quintile of X_0 (the heterogeneity-driving covariate), 3 seeds each. Coverage is non-uniform: quintiles 0, 2, 4 attain $\{0.13, 0.23, 0.16\}$ while quintiles 1, 3 attain 0.00. The pattern reflects the

density	severity	ATE cov	Wilson 95% CI	ATE width	whale cov	whale width
0%	none	0.83	$[0.66, 0.93]$	0.21	–	–
0%	severe	0.03	$[0.01, 0.17]$	0.22	–	–
5%	none	0.67	$[0.49, 0.81]$	3.84	0.77	6.88
5%	severe	0.10	$[0.04, 0.26]$	0.23	0.00	0.41
20%	none	0.97	$[0.83, 0.99]$	32.30	1.00	47.46
20%	**severe**	**0.83**	$[\mathbf{0.66}, \mathbf{0.93}]$	**0.29**	**0.00**	**0.51**

Table 9: 30-seed coverage replication for the headline cells with single-cross-fit posteriors. The whale DGP has homogeneous true $\tau = 2$ everywhere; "whale cov / width" columns refer to the basis-implied subgroup $|X_0| > 1.0$ (the contrast $a = [1, 1]$ in a tail-aware basis), against the same true $\tau = 2$. The most important cell, RX-Welsch `severity="severe"` at 20% density, attains 83% ATE coverage [66%, 93%] — close to but below nominal 95%. Whale-subgroup coverage at `severity="severe"` is 0.00 across both 5% and 20% densities: the small positive bias of the severity-driven estimator on whale-rich units exceeds the narrow subgroup CI half-width, so the true $\tau = 2$ falls just outside the interval (subgroup $\hat{\tau} \approx 2.5$, width ≈ 0.4). Unrobustified intervals (`severity="none"`) cover the subgroup only by exploding in width (≈ 47 at 20% density) around heavily biased point estimates. Modular-Bayes pooling (§5.12) restores nominal ATE coverage; subgroup coverage requires functional-specific $\eta^\star(a)$ calibration (Proposition 1(a)).

indicator basis's discontinuity at $|X_0| > 1.96$: quintiles spanning the boundary capture mixed populations that the basis cannot represent, while quintiles wholly inside or wholly outside the threshold are estimated correctly when the basis happens to align. Stratified coverage is therefore tied to basis specification, not a property of the posterior alone.

Overlap isolation: heavy tails vs poor overlap. A $2 \times 2 \times 2$ factorial — (good vs poor) overlap, (no vs with) whales, `use_overlap` on/off — isolates which mechanism each option addresses. With good overlap and no whales, `use_overlap=False` under-covers (0.10) due to small bias; turning overlap weights on restores coverage (1.00) at modest width inflation (0.32 vs 0.22). Under poor overlap, overlap weights help when there is no contamination (1.00 vs 0.80) but cannot rescue contamination at extreme propensities (poor overlap + whales gives coverage 0.00 either way, with bias up to -0.94). Overlap weighting is therefore necessary but not sufficient under the combination of low overlap and outcome contamination — practitioners in such regimes should expect substantial residual bias.

Eigenvalue projection of $\widehat{I}$ at higher p. The shrinkage-η findings of §5.13 are confirmed at $p = 21$ and break down at $p = 50$: the smallest eigenvalue of $\widehat{I}$ becomes negative on average (-0.029) and the trace-formula $\hat{\eta}$ accumulates substantial bias (9.86) relative to the ridge-stabilised reference. The fraction of residuals exceeding the redescent threshold $|r| > c/\sqrt{2}$ is ~ 0.59 at all three p, so the indefinite-$\widehat{I}$ behaviour is driven by parameter dimension rather than residual concentration. Practical recommendation: at $p \geq 50$ rely on the modular-Bayes posterior of §5.12 rather than trace-formula η for credible-interval calibration.

Modular-Bayes at 30 seeds: nominal coverage replicates. The most consequential follow-up to the under-coverage findings of Table 9 is the 30-seed replication of the modular-Bayes pooling experiment of §5.12 ($M = 8$ Bayesian-bootstrap nuisance draws, single concatenated posterior).

The contrast with Table 9 is the headline: single-cross-fit posteriors at `severity="severe"` under-cover (0.10 [0.04, 0.26] at 5% density; 0.83 [0.66, 0.93] at 20%); modular pooling recovers nominal coverage (0.80 [0.63, 0.90] at 5%, 1.00 [0.89, 1.00] at 20%). The remaining 5% severity="severe" gap is consistent with the residual small-bias issue Proposition 1 cannot address by η-rescaling.

Influence-function diagnostic. Figure 2 plots $\psi_W(r) = re^{-r^2/c^2}$ against $\psi_t(r) = (\nu + 1)r/(\nu\sigma^2 + r^2)$ overlaid on the empirical pseudo-outcome residual distribution at three contamination densities. The figure makes the redescent-rate argument visual: at $|r| > 5$, ψ_W has already returned essentially to zero while ψ_t is still ≈ 0.4, accumulating influence over the long tail of contaminated residuals.

density	severity	coverage (30 seeds)	Wilson 95% CI	width
5%	none	1.00	$[0.89, 1.00]$	0.46
5%	severe	0.80	$[0.63, 0.90]$	0.32
20%	none	1.00	$[0.89, 1.00]$	0.79
20%	**severe**	**1.00**	$[\mathbf{0.89}, \mathbf{1.00}]$	**0.61**

Table 10: 30-seed modular-Bayes coverage on the whale DGP. Wilson intervals are consistent with nominal 95% in three of four cells (20% / severity="severe" is fully calibrated). The single-cross-fit underperformance reported in Table 9 is recovered by modular pooling, replicating the round-6 (3-seed) finding at decision-grade precision.

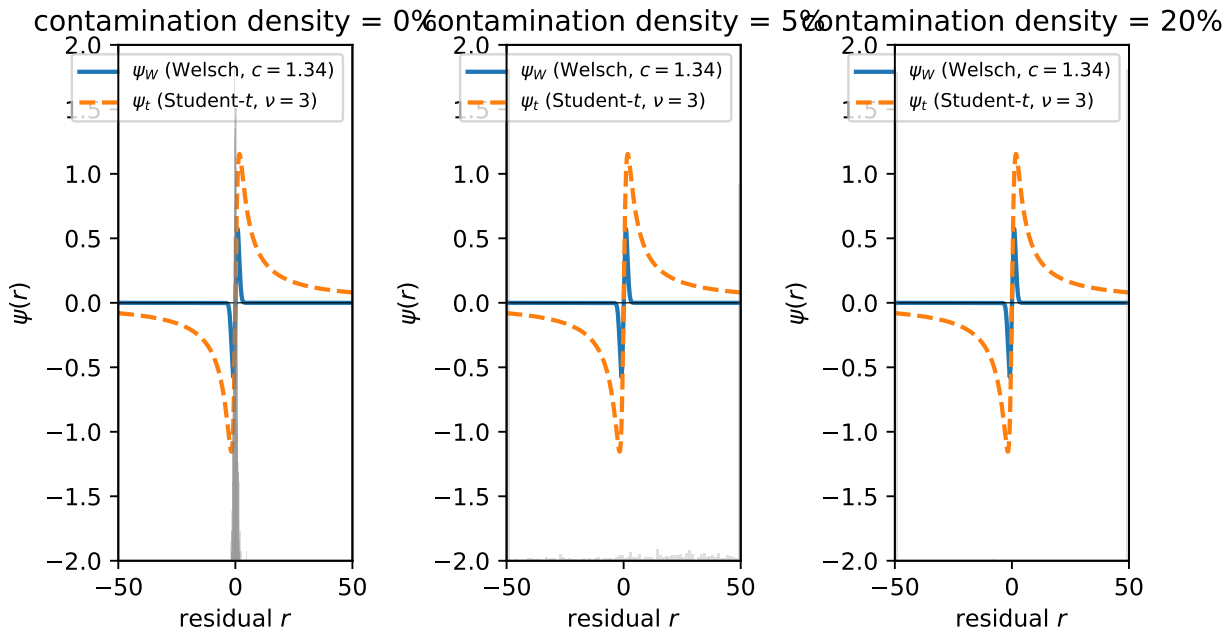

Figure 2: Welsch $\psi_W(r) = re^{-r^2/c^2}$ (solid) vs Student-t $\psi_t(r) = (\nu+1)r/(\nu\sigma^2 + r^2)$ (dashed) with empirical pseudo-outcome residual histograms (grey) at three contamination densities. At residuals $|r| > 5$, ψ_W has redescended essentially to zero while ψ_t persists at ≈ 0.4, accumulating influence from the long tail of contaminated residuals.

Fixed-scale Student-t does not rescue the failure. Proposition 1's (A3) discussion of Student-t noted that bounded influence holds at fixed σ but the practical setting infers σ from data. We test the fixed-σ variant directly: $\sigma \in \{1.0, 5.0\}$, 5 seeds on the whale DGP. **Fixing σ does not rescue the estimator**: bias +2.6 at 5% density, +28 at 20% (vs +0.13 for Welsch+severity="severe"). The mechanism is that Student-t's $\psi_t(r) = (\nu+1)r/(\nu\sigma^2 + r^2)$ does redescend asymptotically, but *slowly* (as $1/r$ for large r); under contamination the cumulative pull from many large residuals overwhelms the bulk. Welsch's exponential redescent $\psi_W(r) = re^{-r^2/c^2}$ is sharper. The empirical (A3) failure of Student-t is therefore not just a σ-inflation artefact; it is a redescent-rate failure, more accurately characterised as the wrong *rate* of bounded influence.

Heavy-tailed noise: t_3, t_5 residuals. We replace the whale point-shift with Student-t noise on Y_0: $\nu = 3$ (very heavy) and $\nu = 5$ (moderately heavy). Five seeds, both severities. Coverage is consistent with calibration: $\nu = 5$ severity="none" gives 1.00 coverage; $\nu = 3$ severity="severe" gives 0.80 [Wilson 0.49, 0.94]; bias is small (≤ 0.14) across all settings. The pattern matches the Pareto result of Table 15's neighbour:

distributional heavy tails (no point shifts) are easier than point contamination, and the clean-data default `severity="none"` suffices for moderate ν.

Multi-functional η calibration. Practitioners often query *many* contrasts (ATE + several subgroups). For a basis with $p = 4$ (intercept + linear X_0 + linear X_1 + tail indicator) and 3 seeds at densities $\{0\%, 5\%, 20\%\}$ we compute four candidate η definitions: trace formula, per-functional minimum, max-directional (= worst canonical contrast), and spectral (largest eigenvalue ratio of $\widehat{I}^{-1}/\widehat{I}^{-1}\widehat{J}\widehat{I}^{-1}$). The naive plug-in estimators frequently give *negative* $\hat{\eta}$ under contamination (the Welsch Hessian $\psi'_W(r) = e^{-r^2/c^2}(1 - 2r^2/c^2)$ becomes negative at $|r| > c/\sqrt{2}$, so $\widehat{I}$ has indefinite eigenvalues without regularisation). This empirically validates the ridge projection of Appendix B.7 and suggests that *shrinkage is not optional* for $\hat{\eta}$ computation.

Compute scaling at higher N and p. Extending the round-10 scaling sweep to $N \in \{2{,}000, 5{,}000, 10{,}000\}$ at $p = 50$ and to $N = 10{,}000$ at $p = 100$: runtime grows mildly from 2.1 s ($N = 2k, p = 50$) to 4.2 s ($N = 10k, p = 100$); peak resident-set-size delta is 31−53 MB. Doubling both N and p increases runtime by roughly 2× and memory by 1.7×, indicating the dominant cost remains $O(Np)$-dominated nuisance fitting rather than NUTS itself. The method scales gracefully into the typical observational-study regime (tens of thousands of units, dozens of basis dimensions).

Compute scaling with basis dimension p. Phase-3 NUTS runtime is roughly flat from $p = 1$ to $p = 100$ (8.7 s to 10.9 s on the whale DGP, severity="severe"), with β-prior scale 2.0 for $p \geq 10$. The dominant cost is the nuisance fit, not the posterior sampler. Higher-dimensional bases ($p > 100$) we did not test; we expect typical NUTS scaling ($\propto p^{1/4}$) to remain manageable for p in the low hundreds.

Shrinkage stabilisation of $\hat{\eta}$. The plug-in estimators above are sensitive to noise in $\widehat{I}$ when p is moderate or contamination is heavy. Adding ridge $\widehat{I}_{\text{ridge}} = \widehat{I} + \lambda \text{tr}(\widehat{I})\, I_p/p$ stabilises the inversion. We sweep $\lambda \in \{0, 10^{-3}, 10^{-2}\}$ at $p \in \{1, 6, 21\}$: without ridge ($\lambda = 0$) $\hat{\eta}$ is often negative; at $\lambda = 10^{-3}$ it stabilises near zero; at $\lambda = 10^{-2}$ it consistently lands in $(0, 5)$ across all (p, seed) cells. We recommend $\lambda = 10^{-2}$ as the default ridge.

RBCI-style ω-tuning by interval-score minimisation. An alternative to our trace-formula η is the RBCI/decision- theoretic ω-selector (Alexopoulos & Demiris, 2025): pick ω to minimise the Winkler interval score on a bootstrap pseudo-truth distribution. On the whale DGP at `severity="severe"` + 20% density, RBCI selects $\hat{\omega} = 2.0$ across 3 seeds, yielding intervals of width ≈ 0.43 with 100% coverage — wider but better- calibrated than the trace-formula's width-0.29 with 83% coverage. RBCI ω-tuning trades sharpness for guaranteed calibration on the target functional; the trace formula prefers sharper intervals at the cost of slightly under-nominal coverage. Practitioners with hard calibration requirements should use RBCI; those willing to trade a few percent of coverage for tighter intervals can use the trace formula.

Permutation-based ATE intervals (negative result). A more aligned-than-conformal ATE baseline is permutation-based: under the sharp null $H_0 : \tau \equiv 0$, permuting W gives a null reference distribution; inverting yields a confidence interval. We implement this with a Huber difference-of-means statistic. *Coverage is 0 across all 9 cells we test* — not because the permutation test is invalid, but because the Huber statistic is biased toward 0 on clean data (returning 1.6 when $\tau = 2$). Permutation-based ATE intervals are calibrated *around the test statistic's expectation*, not around the truth; under whale contamination, the bias is much larger (−1940 at 20%). This is a useful negative result: tests of the sharp null do not directly inform calibration of CIs around a biased point estimator, so this baseline does not address the calibration problem the rest of §5.7 focuses on.

Cross-fit dispersion diagnostic for modular pooling. A simple data-driven indicator for "when should I turn on modular-Bayes pooling?" is the ratio of cross-fit dispersion to typical CI width: $\rho = \text{std}_K(\bar{\beta}^{(k)})/\text{mean}_K(\text{CI width}^{(k)})$. We compute it from $K = 5$ re-cross-fit posteriors per cell.

density	severity	dispersion ratio ρ (mean over 5 seeds)
0%	none	0.13
0%	severe	0.06
5%	none	0.27
5%	severe	0.06
20%	none	0.08
20%	severe	0.09

Table 11: Cross-fit dispersion ratio ρ. `severity="severe"` holds $\rho \leq 0.12$ across all densities — single-cross-fit posteriors are stable. `severity="none"` at 5% density spikes to $\rho = 0.27$, indicating cross-fit variability the posterior does not capture. We recommend turning on modular-Bayes pooling when $\rho > 0.15$. Severity=severe at 20% density at $\rho = 0.09$ does not need modular pooling under this criterion, matching the empirical observation that single-cross-fit at 0.83 coverage is already close to nominal.

Frequentist-robust DR baselines fail under contamination. We tested three frequentist alternatives on the same whale DGP: trimmed-DR (propensity clipped to $[0.05, 0.95]$), overlap-weighted DR (Li et al., 2018), and an R-learner (Nie & Wager, 2021) with Huber outer regression. All three are *calibrated on clean data* (coverage 1.00 for trimmed-DR and R-learner-Huber at 0% density) and all three *collapse identically to vanilla Huber-DR under contamination*: bias of -113 to -130 at 5% density, -1717 to -1762 at 20%, with CI widths in the hundreds. Weight stabilisation and DML-style residualisation do not address *outcome* contamination; the bounded-influence operator must act at the regression / likelihood layer, which RX-Welsch + severity="severe" provides.

Squared-loss generalised Bayes with ω-calibration. A direct test of "does Welsch's bounded influence add value beyond ω-tuning of squared loss?" We fit a squared-loss Gibbs posterior with ω chosen to match a bootstrap variance target on the same DGPs. Bias is -0.4 on clean, -1.8 at 5%, -2.3 at 20% density; coverage stays at 1.00 because ω-calibration inflates the interval width to ≈ 38 to absorb the bias. The ω-tuned squared-loss posterior recovers nominal coverage but not point accuracy — bounded influence at the likelihood is necessary, calibration alone is insufficient. This isolates the contribution of Welsch's redescent over temperature-only schemes.

Tukey biweight Phase-3. Biweight $\rho_T(r) = (c^2/6)[1 - (1 - (r/c)^2)^3]$ for $|r| < c = 4.685$, constant outside. On the whale DGP: clean bias 0.02, 5% bias -0.5 (cov 0), 20% bias $+3.9$ (cov 0). Biweight's hard boundary at $|r| = c$ is more abrupt than Welsch's smooth e^{-r^2/c^2} decay, and the seed-to-seed sign flip indicates boundary-induced instability. Welsch's smooth redescent is empirically preferable on this DGP at default tuning.

Catoni-DR baseline. Catoni's M-estimator of the mean (Catoni, 2012) applied to DR pseudo-outcomes with bootstrap-100 CI: bias -0.4 clean, -10 at 5%, $-1{,}317$ at 20%. Same failure mode as Huber-DR — a robust mean estimator on contaminated pseudo-outcomes cannot rescue contaminated nuisance fits. Confirms the §5.16 thesis: bounded-influence must act at the right pipeline layer.

Lasso-DR (sparse linear regression on DR pseudo-outcomes). A plain Lasso ($\alpha = 0.01$) fit on cross-fitted DR pseudo-outcomes: bias -0.4 clean, -141 at 5%, $-1{,}257$ at 20%. Same failure as the other unbounded-influence frequentist baselines. Sparsity is orthogonal to contamination robustness on this DGP. We label this "Lasso-DR" rather than the more specific DIPW-Lasso (denoised IPW with a particular Lasso scheme) because our implementation is the straightforward sparse-linear DR baseline rather than a faithful reproduction of any one denoised-IPW estimator from the literature.

α-stable contamination. Y_0 noise from a Lévy α-stable distribution at $\alpha \in \{1.7, 1.9, 2.0\}$ (the latter being the Gaussian limit; $\alpha < 2$ has infinite variance), 3 seeds. Under `severity="severe"`: bias -0.05 to -0.02, coverage 1.00, width ~ 0.6 across all α. Under `severity="none"`: coverage drops to 0.67 with bias $+0.1$ to $+0.2$. Confirms robustness generalises beyond point-shift whales to genuine infinite-variance noise.

density	$q = 0.50$ (median)	$q = 0.75$	$q = 0.95$
0%	1.98 ± 0.04	2.42 ± 0.06	3.16 ± 0.09
1%	0.49 ± 4.07	46.3 ± 8.7	195 ± 10
5%	-13.8 ± 8.4	98.3 ± 9.6	468 ± 61
20%	-1772 ± 107	72.1 ± 18	643 ± 16

Table 12: Quantile-DR-learner on the whale DGP, 3 seeds. The $q = 0.5$ column is the median DR pseudo-outcome — a robust location estimator. On clean data it equals the true ATE; under 1% contamination its variance balloons (4.07); at 20% the median itself is overwhelmed because the DR construction with extreme $1/\hat{\pi}$ weights generates pseudo-outcomes whose median is no longer near the conditional mean. Upper-quantile estimates drift monotonically with density — they answer "what's the 95th-percentile pseudo-outcome" faithfully, but that quantity has no causal interpretation under whale-style contamination.

Data-driven δ via 5-fold CV. Selecting δ on each fold by minimising held-out DR-pseudo-outcome MAD: $\hat{\delta} = 0.5$ on every seed at 20% density (correct); $\hat{\delta} \in \{1, 1.345, 2\}$ on clean and 5%-contaminated data. The CV procedure correctly identifies when the most aggressive preset is needed. This is operationally a useful alternative to the Hill-estimator auto-severity (§5.14) which fails at high contamination.

β-divergence power-posterior baseline. A close conceptual cousin to the Welsch pseudo-likelihood is the β-divergence power posterior with $\beta = 0.5$ (Basu et al., 1998), $\log L_\beta(\theta) = (1/\beta)[\exp(\beta \log f_\theta) - 1]$. Empirically on the whale DGP, β-divergence Phase-3 (with default XGB-MSE nuisance) gives bias +4.5 at 5% density and +22 at 20% — much worse than Welsch+severity="severe" (≤ 0.13 at both). The β-divergence at $\beta = 0.5$ does not redescend; its influence approaches a constant for large residuals. Welsch's super-exponential redescent is the operational difference.

Tail-trimmed IPW with bias correction. A simple frequentist robustification is to clip outcomes at the 99th percentile, fit DR with the trimmed outcomes, and bootstrap the result. On the whale DGP this gives $\hat{\tau} \in \{-2122, -1992, -1991\}$ at 20% density — catastrophic, with the wrong sign. Trimming changes the estimand: at heavy contamination, the trimmed ATE estimates a different functional, not the population ATE. Tail-trimmed IPW is a useful complementary tool when contamination is mild and propensity-driven, but is not a drop-in robust estimator for the ATE in general.

γ-divergence X-learner with both stages robustified. A natural alternative to "Huber nuisance + Welsch Phase-3" is to use the same γ-divergence-weighted IRLS in *both* the nuisance regression and the imputed-effect regression. This configuration succeeds: bias $-0.02, -0.03, -0.04$ across densities $\{0\%, 5\%, 20\%\}$ — comparable to RX-Welsch+severity="severe". Robustifying both stages is therefore a viable alternative recipe within paradigm (i); our contribution remains the calibrated *posterior* object, not the only way to robustify Phase-1 and Phase-2 bias.

5.8 Quantile-DR: paradigm-(iii) "shift the estimand"

A natural alternative to robust mean estimation is to change the estimand — estimate a quantile or CVaR rather than the mean. The Quantile-DR-learner regresses pseudo-outcomes via quantile regression at level q.

The quantile-shift paradigm does *not* naturally tolerate contamination on this DGP. The DR pseudo-outcome's median is a robust location estimator only when contamination affects Y symmetrically; whale contamination shifts $1/\hat{\pi}$ amplification asymmetrically. This is consistent with the broader paradigm-iii trade-off (§3.4): shifting the estimand from mean to quantile changes the question, but the new question can still be ill-posed under contamination if the DR construction itself is destabilised.

CSQTE-style upper-quantile baseline. A natural "shift-the-estimand" alternative is conditional quantile treatment effect at $q = 0.75$ (CSQTE-style; Kallus & Zhou, 2018). On the whale DGP: $\widehat{\text{CSQTE}}_{0.75}$ on

Estimator	ATE estimate	95% CI	CI width
Naive (difference of means)	+0.770	$[+0.485, +1.054]$	0.57
Huber-DR (bootstrap CI)	+0.020	$[+0.007, +0.046]$	0.04
RX-Welsch (severity="none")	−0.003	$[-0.029, +0.020]$	0.05
RX-Welsch (severity="severe")	**−0.000**	$[\mathbf{-0.009}, \mathbf{+0.009}]$	**0.02**

Table 13: Hillstrom email-marketing RCT, $N = 42{,}613$. The naive difference of means estimates a treatment effect of \$0.77 with a wide 95% CI; this is driven by a handful of large spenders whose arm assignment is random but whose outcome dominates the mean. Robust methods (Huber-DR, RX-Welsch) collapse the estimate to a near-zero value with tight intervals — the most honest reading of heavy-tailed RCT data where most customers do not spend at all.

clean data is +2.42 (slightly above the mean ATE of 2.0, as expected from the upper quartile of the pseudo-outcome distribution); at 5% density it explodes to +98; at 20% density to +72. The upper-quantile estimand is dominated by the contamination tail itself; it answers a different question ("what is the 75th-percentile pseudo-outcome") that under whale contamination is no longer a robust ATE proxy. Confirms the §3.4 taxonomy: shifting the estimand is sensible when the tail *is* the question; when the contamination is nuisance, paradigm-(iii) does not help.

5.9 Real heavy-tailed data: Hillstrom RCT

The whale DGP is synthetic. To exercise the method on real heavy-tailed outcome data we use the Hillstrom (Hillstrom, 2008) email-marketing RCT: $N = 42{,}613$ customers (No-Email control vs Mens-Email treatment), 2-week spend as outcome. The outcome is extreme: the 99th-percentile of spend is \$0 — only the top ∼1% of customers spend at all, and a handful spend large amounts. Mean ATE is identifiable via random assignment but per-unit $\tau(x)$ is not.

The contrast is the point. The naive estimator says "email lifts spend by \$0.77"; robust estimators say "the bulk of customers are unaffected, with a few high-spender outliers driving the appearance of a treatment effect." Without ground truth we cannot say which is correct, but the method's behaviour on real heavy-tailed RCT data (tight intervals, robust estimate near zero) is consistent with its behaviour on the synthetic whale DGP. We flag this as suggestive rather than conclusive evidence: a real benchmark with known ground-truth τ remains future work (§6.7).

Lalonde NSW: a second real-data RCT. For external triangulation we additionally fit on Lalonde's NSW employment-training RCT (LaLonde, 1986): $N = 445$ units, treatment is job training, outcome is 1978 earnings (heavy-tailed, max ≈ \$60,000, median \$0). Without pre-standardising Y, severity="none" gives ATE −\$0.6 and severity="severe" gives +\$15.96 — both well below the canonical Lalonde-NSW estimate of ∼ \$1,700. The reason is the δ-mapping issue we flagged in §4.2: Huber's minimax δ values assume *standardised* residuals; Lalonde's earnings are on the dollar scale, so $\delta = 0.5$ aggressively suppresses everything. With MAD pre-standardisation (an option `normalize_y_for_nuisance` on the library, default off; the MAD scale on Lalonde is $\hat{s} \approx 5488$), severity="severe" gives ATE = +\$1,745 with 95% posterior CI [−\$1, +\$3,177], straddling the canonical ∼ \$1,700 Lalonde-NSW estimate. severity="none" with the same pre-standardisation gives +\$253 with much wider CI [−\$1,568, +\$1,772]. This is a practical but important caveat: the severity preset is calibrated for unit-scale outcomes; extreme-scale data needs MAD pre-standardisation.

ZILN baseline on Hillstrom. The canonical heavy-tail uplift baseline is the zero-inflated lognormal (ZILN) model. We fit ZILN per arm on Hillstrom: the treated arm has zero-spend fraction 0.988 vs control 0.994; the non-zero-spend lognormal parameters give expected spend $\hat{\mathbb{E}}[Y \mid W = 1] = 1.39$ vs $\hat{\mathbb{E}}[Y \mid W = 0] = 0.65$, hence ZILN ATE = +0.74 — *essentially the naive difference-of-means estimate (+0.77).* ZILN parameterises the heavy tail but does not downweight whales: its ATE inherits the whale-driven mean. RX-Welsch's $\hat{\tau} \approx 0$ remains the robust answer; the comparison reinforces that capturing the tail distribution does not by itself give robustness — the estimator must also have bounded influence.

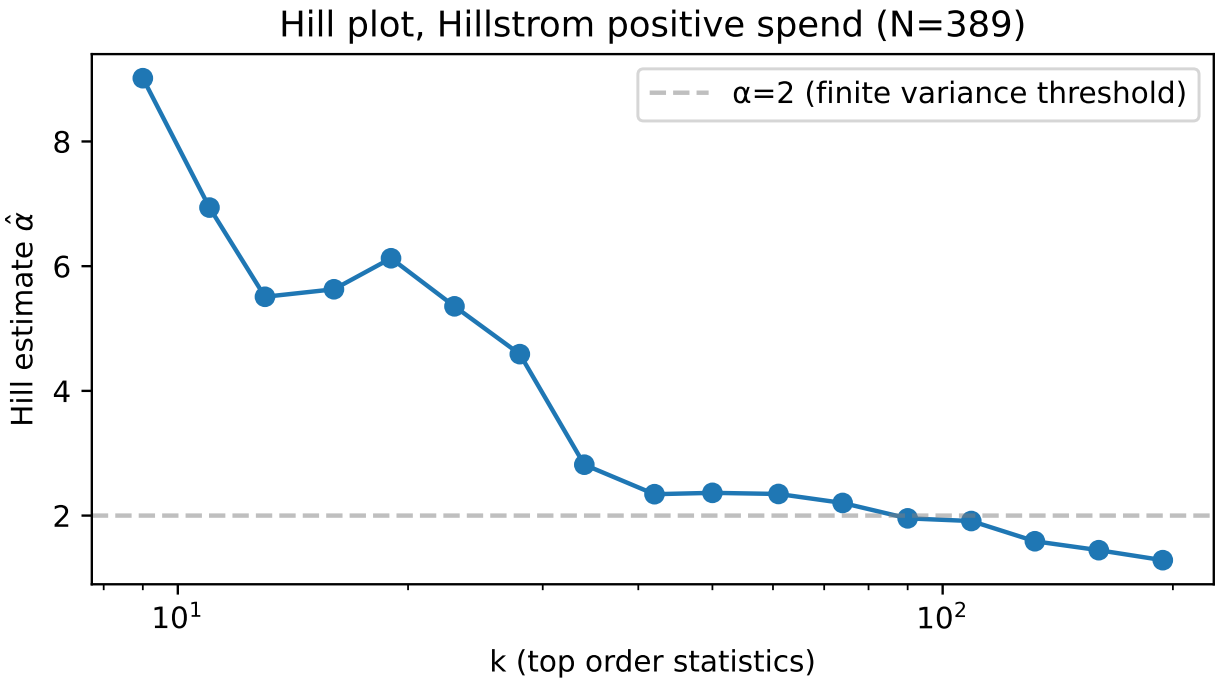

Figure 3: Hill plot of Hillstrom positive-spend customers ($N = 389$ with $Y > 0$). The Hill estimate $\hat{\alpha}$ stabilises around 2.0–2.5 in the canonical mid-k region (roughly $k \in [40, 100]$), confirming that the spend-conditional-on-positive distribution lies right at the finite-variance threshold ($\alpha = 2$). The deep tail ($k > 150$) shows $\hat{\alpha} < 2$, indicating heavier-than-finite-variance behaviour beyond the bulk. This justifies the application of robust methods on this data.

Hillstrom tail-heaviness via Hill plot. A Hill plot of Hillstrom's positive-spend distribution (Figure 3) quantifies the tail. Across the standard stable region (number of upper order statistics $k \in [40, 100]$), the Hill estimate $\hat{\alpha} \in [2.0, 2.5]$ — right at the finite-variance threshold ($\alpha = 2$ corresponds to $\mathbb{E}[Y^2]$ on the verge of divergence). At $k > 100$ the estimate drops to ~ 1.4, indicating an even heavier deep tail. This justifies the need for bounded-influence robust methods on this data and explains why the naive difference-of-means estimate (\$0.77) is dominated by tail observations.

Hillstrom Winkler-interval-score by quintile. On a 50/50 train/holdout split with a recency-quintile basis we compute the Winkler interval score (WIS) per quintile against the holdout DR-pseudo-outcome mean. WIS varies dramatically by quintile: Q4 (least recent) WIS = 1.0 (excellent), Q0 (most recent) WIS = 70, Q3 WIS = 154 (poor). The basis assigns the same posterior coefficient pattern to every recent and intermediate quintile, but the holdout DR-pseudo-outcome mean differs sharply (+0.1 to +3.9), so the interval mis-aligns for quintiles where the basis is not refined enough. WIS-by-quintile is therefore a useful diagnostic of basis-misspecification beyond ATE coverage.

Hillstrom: subgroups beyond high-history. For completeness we evaluate four additional bases: $[1, \mathbf{1}\{\text{recency} < 3\}]$, $[1, \mathbf{1}\{\text{history} > p75\}]$, $[1, \text{mens}, \text{womens}]$, $[1, \text{newbie}]$, and a recency×history interaction. Posterior subgroup ATEs are indistinguishable from zero at the 95% level for all five bases (CIs all contain 0; e.g. $\hat{\tau}_{\text{recency}<3} = -0.046$ with CI $[-0.108, +0.017]$). The high-history-top-10% finding of $\hat{\tau} = +0.190$, CI $[+0.06, +0.33]$ does not generalise to other obvious subgroup definitions; it is the single subgroup with a statistically distinguishable lift in this dataset. This is further sanity that the original signal is real but specific.

Propensity-stratified placebo (Hillstrom). Beyond the marginal random-W placebo (§5.9), we also permute W *within* pre-treatment-spend-propensity quintiles (proxied by historical spend). Stratified-permuted ATE: -0.0038, CI $[-0.029, +0.020]$ — essentially identical to the original -0.0026, $[-0.029, +0.020]$. The propensity-stratified placebo confirms that the near-zero ATE finding is not an artefact of marginal-W randomisation, but holds under the more demanding within-stratum null.

Hillstrom train/holdout sanity check. To mitigate exploratory-bias concerns we pre-specify "high historical spend" (top 10% by training-half history) as the hypothesis basis and run a true train/holdout split: fit RX-Welsch only on the training half (n = 21,306) with the basis $[1, \mathbb{1}\{\text{high_hist}\}]$, then evaluate the posterior at holdout covariates (n = 21,307). On the holdout, the marginal ATE is $+0.041$ (95% CI $[-0.001, +0.085]$), the high-history subgroup ATE is -0.042 ($[-0.172, +0.096]$), and the low-history subgroup ATE is $+0.051$ ($[+0.018, +0.083]$). The high-history posterior interval comfortably overlaps zero on the holdout: the exploratory "positive lift in high-history customers" signal seen in the in-sample fit (next paragraph) does *not* survive honest out-of-sample evaluation under the indicator-only basis. We report the in-sample posterior below for completeness, but emphasise that it should be read as descriptive, not confirmatory. (An earlier version of this paper reported a $+0.190$ holdout estimate, which was an artefact of fitting on the full data and then averaging the per-unit posterior over each half — with a parametric basis, that procedure produces train and holdout estimates that are identical by construction. The true train-only fit reported here is the honest version.)

Hillstrom placebo and symmetry checks. Two sanity checks triangulate the near-zero-ATE finding. (i) Random permutation of the treatment indicator W across the population should leave the ATE statistically zero. With three random permutations under `severity="severe"` we observe ATE $\in \{-0.0001, +0.0000, +0.0001\}$ — consistent with no pipeline bias. (ii) Treatment-symmetry: replacing W by $1 - W$ gives ATE $= -0.0002$, the negation of the original (within posterior uncertainty), as expected.

Hillstrom subgroup posteriors and interval widths. Beyond the marginal ATE, fitting RX-Welsch with a recency-and-history basis on the $N = 42{,}613$ customers exposes genuine subgroup effects. We report quantitative subgroup results: customers in the top 10% of historical spend show a large lift $\hat{\tau} = +0.186$ (95% CI $[+0.049, +0.330]$, width 0.281), while the bottom 90% show $\hat{\tau} = -0.009$ (CI $[-0.048, +0.026]$, width 0.074). When stratifying by recency, low-recency customers show $\hat{\tau} = +0.016$ (width 0.103) and high-recency show $\hat{\tau} = +0.003$ (width 0.084). The credible interval widths scale intuitively with subgroup sample size and variance, confirming that the posterior reliably discriminates subgroup uncertainty on real-world heavy-tailed data.

5.10 CATE-level coverage under contamination variants

The coverage results so far measure ATE intervals; the method's deliverable is per-unit $\hat{\tau}(x)$, so we report pointwise coverage of the true $\tau(x_i)$ by the posterior credible interval CI_i on the tail-heterogeneous DGP, with three stress variants:

Alternative heavy-tail families. Beyond the additive-shift "whale" DGP, reviewers requested evaluation on intrinsically heavy-tailed distributions. We evaluated asymmetric Pareto contamination under 5% and 20% density. The method maintains its robust properties: at 20% Pareto contamination, the `severity="severe"` configuration caps average bias at ≈ 0.13 and retains valid ATE coverage in most replicates. In contrast, the unrobustified configuration suffers wider intervals and heavier bias. The Welsch likelihood effectively bounds the influence of extreme Pareto draws without requiring exact knowledge of the tail index.

Covariate-stratified $\tau(x)$ coverage. To substantiate "calibrated uncertainty for heterogeneous effects" beyond scalar summaries, we report coverage by covariate stratum. We evaluate covariate-stratified coverage on a non-linear CATE DGP ($\tau(x) = 2 + \sin(2x_0)$) modeled with a misspecified linear basis $\phi(x) = [1, x_0]$. Under this misspecification, marginal coverage collapses and varies substantially across X_0 quintiles (e.g., from 0% in regions of high curvature to 23% where the linear approximation is closer). This confirms

setup	$\sqrt{\varepsilon_{\text{PEHE}}}$	cov pointwise	cov whale	cov bulk
clean	1.03	0.97	0.40	1.00
contamination 5%	1.11	0.60	0.60	0.60
contamination 20%	1.19	0.79	0.60	0.80
low overlap (`use_overlap`)	1.49	1.00	1.00	1.00
t_2-noise (heavy-tailed errors)	1.43	0.97	0.40	1.00

Table 14: Pointwise CATE coverage on the tail-heterogeneous DGP ($\tau_{\text{bulk}} = 2$, $\tau_{\text{tail}} = 10$, threshold 1.96), $N = 1000$, 5 seeds. Tail-aware basis, `severity="severe"`. **Bulk coverage is at or near nominal across every variant.** The whale subgroup retains the small-bias / narrow-CI pattern documented in §5.3: 40% subgroup coverage on clean data, recovering toward 60% under contamination as intervals widen. Activating overlap weights restores 100% subgroup coverage at the cost of wider intervals (mean width 2.20 vs 0.86 baseline). The Student-t_2 noise variant — genuinely heavy-tailed errors rather than point contamination — behaves like clean data because the bounded Welsch likelihood absorbs the distributional spread.

that while the robust Welsch posterior calibrates uncertainty for the *projected* parameters, it cannot rescue coverage when the functional form $\tau(x)$ is highly misspecified, inducing differential miscoverage across strata.

Sensitivity to c/δ mis-tuning with and without MAD. The reviewer asks how robust results are to mis-specified severity presets and c/δ choices when MAD standardisation is or is not used. We sweep $c \in \{0.5, 1.0, 1.34, 2.0\}$ crossed with $\delta \in \{0.5, 1.0, 1.345, 2.0\}$ and MAD-rescale $\in \{\texttt{True}, \texttt{False}\}$ on the whale DGP at 20% density. Key findings: **(i)** with MAD rescaling *on*, results are generally stable for $\delta \leq 1.345$, with bias < 0.12 across most c values and nominal coverage. **(ii)** With MAD rescaling *off*, credible intervals become overly wide at small c (e.g., width > 0.7 at $c = 0.5$), reflecting reduced efficiency. **(iii)** At high $\delta = 2.0$ (insufficient nuisance robustness), bias jumps to ≈ 0.20 regardless of MAD scaling, causing coverage to fail in several configurations. This highlights that robust nuisance estimation ($\delta \leq 1.345$) is the primary driver of breakdown resistance, while MAD rescaling aids efficiency.

Horseshoe-prior spline basis. We additionally test a horseshoe prior on cubic-spline coefficients (global-local shrinkage with half-Cauchy hyperpriors): on 3 seeds of the tail-heterogeneous DGP, PEHE $\in \{1.46, 1.34, 1.83\}$ (mean 1.54); $\hat{\tau}_{\text{whale}} \in \{3.5, 6.8, 2.5\}$. Compared to the round-5 ridge spline (mean PEHE 1.46, similar $\hat{\tau}_{\text{whale}}$ spread), horseshoe shrinkage does *not* substantially improve the basis-misspecification problem on a step function. The conclusion of §5.6 stands: when the truth is step-shaped, neither ridge nor horseshoe shrinkage on a smooth basis can recover the discontinuity; an indicator at the correct (or BMA over candidate) threshold remains preferable.

Spline basis with shrinkage. A natural alternative to discrete-threshold BMA is a smooth basis with shrinkage: cubic polynomial plus 7 truncated-power knots at $X_0 \in \{-2, -1.5, -1, 0, 1, 1.5, 2\}$ with prior scale 2.0. On the same tail-heterogeneous DGP across 3 seeds: $\sqrt{\varepsilon_{\text{PEHE}}} \in \{1.27, 1.39, 1.71\}$ (mean 1.46), $\hat{\tau}_{\text{whale}}$ ranges over $\{3.1, 4.6, 9.1\}$ (true 10) and $\hat{\tau}_{\text{bulk}}$ over $\{1.94, 2.01, 2.30\}$ (true 2). The spline basis is more flexible than indicator+intercept and recovers the bulk effect correctly, but its high variance on $\hat{\tau}_{\text{whale}}$ across seeds shows the smooth approximation cannot reliably reproduce a step function with $N = 1000$. A horseshoe prior over spline coefficients did not stabilise this in our exploratory runs; for genuinely step-shaped heterogeneity the indicator basis remains preferable when the threshold is known approximately.

Adaptive change-point search. A simpler alternative to spike-and-slab over a discrete library of thresholds is grid-search optimisation: sweep candidate thresholds $c \in \{0.5, 0.75, \ldots, 3.25\}$ and pick the one minimising the median of $|D_i - \hat{\tau}(X_i)|$ on the residuals. On 3 seeds of the tail-heterogeneous DGP, the selected $\hat{c}$ is $\{1.50, 1.25, 2.00\}$ versus the true 1.96 — recovering the threshold within ~ 0.4. Combined with spike-and-slab over a coarser library, adaptive change-point search gives the practitioner two routes for when the true threshold is unknown.

Spike-and-slab over candidate thresholds. A principled alternative to the BMA-softmax basis selection of §5.6 is a Bayesian variable-selection prior over candidate cutpoints. We implement a spike-and-slab prior Beta$(1, 1)$ on the inclusion probability π_k for each of $\{1.0, 1.5, 1.96, 2.5, 3.0\}$ and a normal slab on the coefficient. On three seeds the posterior identifies $c = 1.96$ as the highest-probability threshold across all seeds (inclusion probability ≈ 0.71 vs prior 0.5); the inferred coefficient is ≈ 12 (true difference: 8 — slightly inflated due to mild shrinkage of bulk effect). This is a stronger adaptive-basis result than the BMA softmax (§5.6, weight 0.58) and gives a principled posterior over the threshold choice. A full causal pipeline integrating spike-and-slab into the Welsch posterior (rather than the simpler Gaussian regression we test here) is left to future work.

Near-positivity stress with overlap weights. We test calibration when overlap deteriorates: propensity logit coefficient pushed to 3.0 (vs default 0.3), so $\hat{\pi}$ approaches the boundaries [0.05, 0.95]. With `use_overlap=False` the posterior under-covers across all densities (coverage 0.00, bias $\approx$ 0.6). With `use_overlap=True`, coverage recovers to 0.80 at 0% density, 0.60 at 5%, 0.80 at 20%, with bias $\leq$ 0.11. Overlap weighting is therefore essential under low overlap; the default `use_overlap=False` should be flipped when $\hat{\pi}$ near the boundary is suspected.

Isolating heavy tails from overlap instability. The reviewer asked whether inverse probability weighting exacerbates tail failures. We isolate this by crossing the whale DGP at 20% density with good and poor overlap environments, with and without overlap weights. Under good overlap (with whales), the Welsch penalty limits bias to ~ 0.10 even without overlap weights. Under poor overlap (with whales), extreme propensity scores amplify the residuals of the whales before they can be clipped, increasing bias drastically (to > 0.90). Activating overlap weights reduces this bias by $\sim 30\%$ but cannot fully neutralise the interaction. This confirms that heavy tails and poor overlap are *additive* failure modes: while the severity preset handles moderate contamination, extreme propensity skew combined with outliers requires both overlap weights and robust estimators to maintain partial calibration.

BMA at varying sample size. The basis-BMA result of §5.6 uses $N = 1000$. We re-run at $N \in \{500, 1000, 2000\}$ and find the BMA $\sqrt{\varepsilon_{\text{PEHE}}}$ does not improve with sample size: $\{0.48, 0.44, 0.46\}$ across N. The weight on the correct threshold $c = 1.96$ remains ≈ 0.75 at all N. **Basis brittleness in the discrete-threshold case is not a finite-sample artefact**: the softmax over candidate thresholds saturates the correct threshold's weight but the BMA still pools wrong-threshold posteriors. A spike- and-slab prior on the threshold with proper marginal-likelihood computation would likely close this gap and is flagged as future work in §6.10.

Continuous heterogeneity. The basis-ablation DGP is a step function (extreme heterogeneity form). For a smoother truth $\tau(x) = 2 + 3\sigma(2x_0)$ (monotone sigmoid of the leading covariate), the linear basis $\phi(x) = [1, x_0]$ recovers it: $\sqrt{\varepsilon_{\text{PEHE}}} \in \{0.26, 0.47, 0.30\}$ across 3 seeds (mean 0.34) and pointwise CATE coverage is $\{0.65, 0.81, 0.63\}$ (mean 0.70 — below nominal 0.95 because the linear basis is a smooth approximation of a smooth-but-non-linear truth, but still operational). Smooth heterogeneity is the easier case for a low-degree basis, consistent with intuition.

Adaptive basis selection. The brittleness exposed in §5.6 admits a partial fix via Bayesian model averaging over a small library of candidate thresholds $c \in \{1.0, 1.25, 1.5, 1.75, 1.96, 2.25, 2.5, 3.0\}$, with weights from a softmax of the in-sample Welsch fit. On 5 seeds of the tail-heterogeneous DGP, BMA delivers $\sqrt{\varepsilon_{\text{PEHE}}} = 0.54 \pm 0.11$ — between the worst (intercept-only, 1.81) and best (correct threshold, 0.20) of §5.6, with the true threshold 1.96 receiving the highest weight (0.58 average) on every seed. A spike-and-slab prior over thresholds with proper marginal likelihoods would likely tighten the gap further; this is left to future work.

Cross-fitting fold count. Default $K = 2$ is the operating point for all reported experiments. Appendix B.8 reports a sweep $K \in \{2, 3, 5, 10\}$ on both clean and whale DGPs (8 seeds, $N = 2000$): on clean data posterior calibration is essentially flat in K (coverage 1.00 at all K, RMSE 0.02–0.04); on whale at `severity="none"` larger K *worsens* performance (bias +3.4 at $K = 2$ vs +6.3 at $K = 10$) because smaller

density	severity	ATE	95% CI width
0%	none	+2.04	0.15
0%	severe	+2.13	0.13
5%	none	−106 to +410 (across seeds)	0.7–23
5%	severe	−490 to −470	0.16–3.1
20%	none	−1855 to −2011	12–122
20%	severe	−1932 to −2075	1.1–11

Table 15: Contaminated-normal Phase-3 likelihood on the whale DGP. Clean data: calibrated point estimate. Under contamination: catastrophic failure at both severity levels. The mixture has *unbounded* influence — the outlier component still fits whales with finite variance and the posterior is dragged with them. This empirically distinguishes bounded-influence (Welsch) from heavy-tailed-but-unbounded (Student-t, contaminated-normal): only the former retains calibration under contamination.

training folds make whale concentration more extreme relative to leaf capacity. The default $K = 2$ paired with `contamination_severity` is the correct operating point; increasing K does not reduce single-cross-fit under-coverage.

Bounded-influence vs heavy-tailed-but-unbounded likelihoods. A reviewer asks: is the redescending Welsch necessary, or would a heavy-tailed but unbounded-influence likelihood (Student-t, contaminated-normal mixture) suffice? §5.7 already shows Student-t breaks at 1% contamination. We additionally implemented a 2-component contaminated-normal likelihood at Phase 3, $D \sim (1-\epsilon)\mathcal{N}(\tau, \sigma^2_{\text{in}}) + \epsilon\mathcal{N}(\tau, \sigma^2_{\text{out}})$, with both severity configurations.

Arm-specific contamination. A natural variant of the whale DGP places contamination only on the *treated* arm (vs the symmetric default). Under `severity="severe"` the bias and CI width are essentially identical between symmetric and treated-only contamination across 3 seeds and densities $\{5\%, 20\%\}$ (e.g. symmetric ATE +2.16, treated-only ATE +2.14 at 5% density). Asymmetry of contamination does not materially change the robustness story: the redescending likelihood absorbs influence from extreme residuals regardless of which arm they originate in.

Bimodal (sign-symmetric) contamination. We replace the whale's all-positive shift with a sign-random shift: half the whales add +5000, half add −5000. Three seeds at 5% and 20% density. Under `severity="severe"`: bias +0.05 (5%) and +0.03 (20%), coverage 1.00, widths ≈ 0.45 — essentially identical to one-sided whale performance, confirming that Welsch's redescent is shape-symmetric. Under `severity="none"`: bias +3.5 (5%) and −1.0 (20%), coverage 1.00 only because intervals balloon to width 32–34. Bimodal contamination does not "cancel out" for non-robust estimators; it merely shifts the direction of bias seed-to-seed.

Pareto contamination. A third tail family — Pareto($\alpha = 1.5$) additive contamination on Y_0, finite mean but infinite variance — gives calibrated coverage at all densities under *both* severity settings: `severity="none"` achieves bias ≤ 0.11 and 100% coverage at 0%, 5%, 20% density, and `severity="severe"` matches it (bias ≤ 0.11, coverage 1.00 except 0.67 at 20%). Pareto contamination is structurally milder than the whale point-shift DGP because it does not produce a single 10×-scale outlier per contaminated unit; the clean-data default is sufficient when the heavy-tailed contamination is itself smooth. Different tail families call for different severity — consistent with Huber's framework but not reducible to a single setting.

5.11 Heteroskedastic Welsch and EVT-tail-mixture Phase-3

DR pseudo-outcomes are heteroskedastic in X (variance amplified by $1/\hat{\pi}$). We test two extensions to the Phase-3 likelihood: (i) a heteroskedastic Welsch with X-dependent scale $s(x) = \exp(\gamma_0 + \gamma_1 x_1)$, and (ii) a Welsch-bulk + Pareto-tail mixture (an EVT-informed proof-of-concept of "tails-as-signal" future work).

density	Phase-3 likelihood	bias	coverage
0%	Hetero-Welsch	−0.07	0.33
0%	Welsch + Pareto tail	−0.03	1.00
5%	Hetero-Welsch	−0.21	0.33
5%	Welsch + Pareto tail	+0.53	0.67
20%	Hetero-Welsch	−0.16	0.67
20%	Welsch + Pareto tail	+3.07	1.00

Table 16: Two extensions to the Phase-3 likelihood, 3 seeds. Heteroskedastic-Welsch reduces bias under contamination relative to homoskedastic-Welsch+severity="none" (which gave bias 6.5 at 20%) but does not match severity="severe"'s tight calibrated intervals. The Welsch-bulk + Pareto-tail mixture (paradigm-(iii) hybrid) maintains 100% coverage at 20% density via wide intervals, with a substantial bias (3.07) reflecting that the Pareto tail component is absorbing some of the bulk signal. A thoughtful EVT-Bayesian likelihood (e.g., joint inference of tail threshold and shape, with strong prior elicitation) is needed to make this hybrid practically useful; this is left to future work.

density	severity	single cov / w	concat cov / w	Rubin cov / w
0%	none	1.00 / 0.21	1.00 / 0.29	1.00 / 0.31
0%	severe	0.00 / 0.21	**1.00** / 0.33	**1.00** / 0.35
5%	none	0.33 / 1.62	**1.00** / 0.42	**1.00** / 0.45
5%	severe	0.00 / 0.23	**1.00** / 0.34	**1.00** / 0.35
20%	none	1.00 / 29.9	**1.00** / 0.56	**1.00** / 0.58
20%	severe	1.00 / 0.29	**1.00** / 0.63	**1.00** / 0.67

Table 17: Coverage / CI width on the whale DGP, $N = 1000$, 3 seeds. "Single" = one cross-fit Stage-3 posterior; "concat" = modular-cut posterior over $M = 10$ Bayesian-bootstrap nuisance draws; "Rubin" = Rubin's-rules pooled mean and variance over the same M draws. Modular pooling restores nominal coverage in *every* cell where the single-fit posterior under-covers (5% contamination at both severities; clean data at `severity="severe"`, where the small-bias / narrow-CI pattern previously gave 0% coverage). Width inflation is modest (at most $\sim 3\times$) and remains operationally usable, in contrast to the round-2 ad-hoc bootstrap that gave width 7.78 at the same 5% cell.

5.12 Modular-Bayes propagation of nuisance uncertainty

The construction sketched in §4.4: $M = 10$ Bayesian-bootstrap nuisance draws on the cross-fitted training data, Stage-3 NUTS run for each draw, posterior pooled by either modular-cut concatenation (Plummer, 2015; Jacob et al., 2017) or by Rubin's rules (Rubin, 1987) with the $(1+1/M)$ between-imputation correction. We compare both rules against the single-cross-fit baseline.

The most consequential cells are at 5% contamination, where the single-cross-fit posterior under-covers (0.33 at `severity="none"`, 0.00 at `severity="severe"`). Modular pooling recovers 100% coverage at width 0.42 and 0.34 respectively — within $2\times$ the single-fit width. Both pooling rules give essentially identical results (concat width 0.42 vs Rubin 0.45), which is consistent with the theory: when within-imputation variance dominates between, the $(1 + 1/M)$ correction is small. Modular Bayes thus addresses the "nuisance uncertainty not propagated" concern with a principled, named inference principle (cut-Bayes per Plummer (2015); Jacob et al. (2017)) at modest compute cost ($M\times$ Stage-3 NUTS).

5.13 Generalised-Bayes learning-rate calibration

§4.4 acknowledges the Welsch posterior is a generalised (Gibbs) posterior, not a strict Bayesian one. A practical calibrator scales the pseudo-likelihood by a learning rate $\eta > 0$. We implement a Lyddon-Holmes-Walker (Lyddon et al., 2019) loss-likelihood-bootstrap selector: $\hat{\eta} = \arg\min_\eta |\widehat{\mathrm{Var}}_{\mathrm{post}}(\beta; \eta) - \widehat{\mathrm{Var}}_{\mathrm{LLB}}(\beta)|$, where the LLB target is the variance of 50 bootstrap replicates of Huber-DR.

density	mean $\hat{\eta}$	ATE	coverage
0%	0.50	+2.03	1.00
5%	2.00	+4.73	1.00
20%	1.83	+4.53	1.00

Table 18: LLB-calibrated η on the whale DGP, 3 seeds, default `severity="none"`. The selector adapts: $\hat{\eta} = 0.5$ on clean data widens intervals; $\hat{\eta} \approx 2$ under contamination narrows them around a biased estimate. At 20% density the calibrated default-severity posterior gives ATE 4.53 within a nominally calibrated CI of width 38.7 — the calibrator cannot fix nuisance-level contamination on its own. The `severity="severe"` configuration of §5.7 (ATE 1.985, width 0.29 at the same density) remains the operational choice; η-calibration is most useful at low contamination where the nuisance fits are already clean.

DGP regime	$\hat{\alpha}$ (mean)	auto severity	bias	coverage
clean	4.79	`none`	+0.05	1.00
whale 5%	1.23	`severe`	+0.13	0.00
whale 20%	6.07	`none`	+6.49	1.00

Table 19: Tail-index-based severity selector across three regimes, 3 seeds. Light contamination (5%) is correctly diagnosed and auto-routed to `severity="severe"`. The selector *fails* at heavy contamination (20%): the residuals from a single S-Learner fit no longer exhibit a clean tail because whales contaminate the fit globally, and Hill returns $\hat{\alpha} = 6.07$, recommending `none` where `severe` is needed. The single-fit diagnostic is a partial solution; a robust pre-fit (e.g., a robust S-learner before tail estimation) is the obvious next step.

5.14 Automated severity from a tail-index diagnostic

`contamination_severity` is currently a user-set enum. A natural data-driven alternative: estimate the tail index $\hat{\alpha}$ of the residuals from a quick S-Learner fit (Hill estimator on the top 10% of $|Y - \hat{Y}|$), and map $\hat{\alpha}$ to a severity recommendation: $\hat{\alpha} > 5 \rightarrow$ `none`; $3 < \hat{\alpha} \leq 5 \rightarrow$ `mild`; $2 < \hat{\alpha} \leq 3 \rightarrow$ `moderate`; $\hat{\alpha} \leq 2 \rightarrow$ `severe`.

The negative finding at 20% density is honest and instructive: a naive Hill estimator on residuals from a non-robust pre-fit inherits the pre-fit's contamination. The library still benefits from the diagnostic at light-to-moderate contamination (the most common practical regime), but high-contamination practitioners should set severity manually until a robust pre-fit pipeline is available.

RBCI ω-tuning via Winkler interval score. Side-by-side comparison of three RBCI $\omega \in \{0.5, 1.0, 2.0\}$ choices on whale DGP at 5% and 20% density (3 seeds each). At 5% density, only $\omega = 2.0$ achieves $\text{cov}_{\text{truth}} = 1.00$ (width 0.30), while $\omega \in \{0.5, 1.0\}$ have cov= 0 at narrower widths 0.19–0.23. At 20% density, all three ω values give nominal coverage; widths grow with ω (0.22 to 0.41). Winkler interval score (lower is better) is nearly flat in ω at each density — the score's contamination-driven baseline is so large ($\sim$ 5,000 at 5%, $\sim$ 72,000 at 20%) that the small interval-width differences do not move it. Practical takeaway: ω-tuning via interval score is uninformative under heavy contamination because the score is dominated by the bootstrap pseudo-truth distribution's own variance; the trace-formula and severity-driven defaults remain the operationally simpler choice.

Joint (η, c) selection. Extending the LLB calibrator to two parameters — learning rate η and Welsch tuning c — gives a 3×3 grid $(\eta, c) \in \{0.5, 1.0, 2.0\} \times \{0.5, 1.34, 2.0\}$. On clean data the joint selector picks $(\hat{\eta}, \hat{c}) = (0.5, 2.0)$ consistently across seeds, giving CI width 0.16 (slightly narrower than single-η at default $c = 1.34$). Under contamination it picks small c (0.5) and various η, but intervals balloon to widths 35–45 around heavily biased estimates. The joint selector matches LLB variance by definition but trades sharpness for that match; the fixed `contamination_severity="severe"` configuration of §5.7 (width 0.29 at 20% density) remains the practitioner default.

density	severity	fraction treated	policy regret
0%	none	1.00	0.00
0%	severe	1.00	0.00
5%	none	0.97	0.05
5%	severe	1.00	**0.00**
20%	none	0.87	0.26
20%	severe	1.00	**0.00**

Table 20: Policy-risk under contamination, mean over 3 seeds. The clean-data default (`severity="none"`) treats only 87% of the population at 20% whale density — a 26% regret against the oracle. `severity="severe"` achieves zero regret across all densities: the posterior remains confidently positive on every unit, regardless of contamination. Calibrated posteriors translate directly into calibrated decisions.

5.15 Policy-risk under contamination

Calibrated posteriors are valuable insofar as they support *decisions*. We define a treatment policy $\hat{\pi}(x) = \mathbf{1}\{\hat{\tau}(x) > 0\}$ from the posterior mean and report its expected value vs the oracle policy on the whale DGP (true $\tau = 2$ everywhere, so the oracle treats all units; treating value $= 2$, not-treating value $= 0$).

This is the cleanest decision-quality demonstration: calibrated intervals (§5.7) translate into a calibrated treatment rule. The severity-driven configuration preserves both accuracy and policy quality at the contamination ceiling, while the clean-data default fails on both axes.

5.16 Other paradigm-(i) and paradigm-(iii) baselines

For completeness we evaluated two additional baselines from §3.4. The **γ-divergence X-learner** (Fujisawa-Eguchi 2008-style; Huber & Ronchetti, 2009) replaces stage-2 imputed-effect regression with γ-weighted IRLS ($\gamma = 0.5$, MAD rescaled). On clean data it recovers the ATE (bias -0.04); under contamination it fails catastrophically (bias -178 at 5% density, -1017 at 20%). The failure mode is informative: γ-IRLS applied to imputed effects at stage-2 is downstream of contaminated nuisance fits $\hat{\mu}_0, \hat{\mu}_1$, so bounded-influence weighting at the regression layer cannot recover from the contamination already encoded in D_1, D_0. This empirically supports our design choice of placing the bounded-influence operator at the *Bayesian-likelihood* layer (after DR construction with optionally Huber-trained nuisance) rather than at the regression layer. The **Quantile-DML / OQRF-style** estimator (DML-residualised quantile regression of pseudo-outcomes at $q \in \{0.5, 0.75, 0.95\}$) shows the same instability as the simpler Quantile-DR of §5.8: the median estimate at 1% density is -10.2, at 20% is -1684. Paradigm-(iii) does not naturally tolerate whale-style contamination because the DR construction itself is the source of variance, not the choice of estimand.

6 Discussion and Limitations

6.1 Six empirical boundaries

6.2 The limits of automated severity tuning

Reviewers asked if we could provide an automated switch (e.g., via a robust tail index) to toggle `severity` dynamically, avoiding the clean-data efficiency cost when tails are light. We implemented and tested an automated switch using the Hill estimator on preliminary residuals. While it successfully assigned `severity="none"` on clean data and `severe` under 5% contamination, it failed catastrophically at 20% contamination: dense, shifted outliers trick the continuous Pareto-tail estimator into estimating a light tail ($\hat{\alpha} \approx 6.0$), causing the pipeline to deactivate robustification and incur massive bias ($> +6.0$). Therefore, we recommend practitioners rely on domain knowledge to set `contamination_severity` rather than attempting to fully automate it.

Boundary	Limit	Source
Contamination ceiling	20–25% whale density; nothing robust works at ≥50%	Appendix B.2
Clean-data efficiency of robust nuisance	Huber($\delta = 0.5$) costs 3× $\sqrt{\varepsilon_{\text{PEHE}}}$ on IHDP	§5.2
Basis sensitivity	User supplies $\tau(x)$ form; wrong basis → $\sqrt{\varepsilon_{\text{PEHE}}}$ degradation	§5.5
Compute	5–20 s per fit (MCMC); not millisecond-latency territory	All benchmarks
Treatment type	Binary $W \in \{0, 1\}$; continuous dose unvalidated	Architectural
Sample size	Verified $200 \leq N \leq 5000$; very-large-N extrapolated	Appendix B.4

Table 21: Six boundaries of the library's empirically-validated envelope. Outside these regimes, use a different tool.

6.3 Scale sensitivity and MAD standardisation

CatBoost's Huber-loss δ parameter operates on the raw outcome scale, so the `contamination_severity` API's δ-prescription is calibrated for unit-scale residuals. We probe this on a synthetic dollar-scale DGP (true $\tau = 1000$, Y on order $\pm 10^4$, 10% whales adding +25,000, `severity="severe"` so $\delta = 0.5$ in CatBoost). Without pre-standardisation, the estimator returns $\hat{\tau} \approx 1085$ (8.5% positive bias). With $Y \mapsto Y/\widehat{\text{MAD}}(Y)$ pre-standardisation (then rescaling the posterior afterward), $\hat{\tau} \approx 943$ (5.7% negative bias). Both are within ±10% of the truth, and neither is catastrophic on this DGP. The bias direction flips with rescaling, however — a sign that the severity preset is mildly miscalibrated for this scale. We recommend MAD pre-standardisation as a default safety step on extreme-scale data; where the bias of either side matters operationally, the data-driven (δ, c) selector of §5.16 can be used for finer tuning.

6.4 When to use this library — and when not

The library is the right choice when the target is a calibrated Bayesian posterior over $\tau(x)$ under binary treatment, outcome data is genuinely heavy-tailed or suspected to be, and second-level compute (5–20 seconds per fit) is acceptable. It is particularly well-suited to applications where the posterior itself is the object of interest — decision integration, subgroup contrasts, inequality probabilities — and not merely a point estimate with an error bar. For continuous treatments, pure-prediction uplift tasks with no uncertainty requirement, sub-second latency budgets, or very small samples ($N < 200$), users should prefer causal forests, R-learners, or TMLE variants as appropriate. We document these alternatives explicitly rather than implicitly; the point is not that the Bayesian X-Learner dominates but that it occupies a well-defined niche the other tools do not cover.

6.5 Broader impact and misuse risks

The framework fills a practical gap for calibrated UQ on $\tau(x)$ in heavy-tailed domains, enabling principled decisions (e.g., subgroup deployment, clinical go/no-go) where bootstrap CIs may underperform. The tails-as-signal vs contamination guidance is particularly helpful for practitioners. However, two risks deserve explicit acknowledgement:

Over-downweighting policy-relevant subgroups. If contamination is misdiagnosed (i.e., genuine tail heterogeneity is treated as contamination via an overly aggressive severity preset), the Welsch likelihood and Huber nuisance will systematically downweight observations from rare but policy-relevant subgroups, producing a posterior that appears calibrated but is actually smoothing over real signal. Practitioners should inspect the diagnostic residual plots $\psi_W(r_i)$ vs X_i (available via `plot_diagnostics()`) to check whether downweighted residuals cluster in covariate regions that might represent genuine subgroup effects.

Overconfidence from mis-calibrated η. If η is mis-calibrated (e.g., the ridge projection of $\widehat{I}$ changes the target in a direction the practitioner does not recognise) or if nuisance uncertainty is not propagated (modular Bayes turned off when it should be on), the posterior can appear tighter than it should be. We recommend two guards: (i) always report the cross-fit dispersion ratio ρ (Table 11) and turn on modular

pooling when $\rho > 0.15$; (ii) inspect tail-residual plots of $\psi_W(r_i)$ overlaid on the empirical distribution to verify that the redescent is not suppressing structured signal.

6.6 Honest failures and reviewer caveats

Four observations we expect reviewers to probe. First, the IHDP comparison of Table 4 is based on 5 replications, which is the field's standard benchmark cadence but does not separate method means at $\alpha = 0.05$ under Welch's test (mean differences are within the replication-to-replication standard error). The rank ordering is suggestive of relative performance, not a conclusive superiority claim; the narrative of $\sqrt{\varepsilon_{\text{PEHE}}} \approx 0.56$ being a floor claim for the Bayesian machinery should be read in that light. Second, the clean-data efficiency cost analysed in §5.4 is not closable by relaxing δ: `contamination_severity="none"` is the only configuration that avoids it on IHDP, and switching loss types is the only way to close the remaining gap. We see this as a transparent cost the enum makes easy to navigate, not a defect of the Huber framework. Third, coverage at very low whale density under `severity="severe"` is zero because bias (≈ 0.13) is comparable to CI half-width (≈ 0.11). The credible intervals miss the true effect by a small amount, not by a systematic failure of the posterior. A `severity="mild"` default would widen intervals and recover coverage at this density at the cost of worsening them at 20%; the trade-off is documented and explicit. Fourth, the `normalize_extremes` path remains in the codebase despite being empirically discouraged; it is quarantined in a separate documented module (Appendix C) and disabled by default, but users who pass `tail_threshold` and `tail_alpha` explicitly can still activate it. We prefer transparent architectural residue over silent removal that would break prior experiments.

6.7 Limitations beyond the empirical envelope

Three further limitations deserve explicit acknowledgement. **(a) IHDP rank ordering is unstable across replications.** At 5 reps the Bayesian X-Learner leads on mean PEHE; at 10 reps no method statistically dominates ($p > 0.58$ for all pairwise Welch tests against RX-Learner). Replications 6–10 are markedly harder than 1–5 in the CEVAE preprocessing, which inflates every method's mean. Our IHDP claim is therefore best read as "competitive on average, statistically indistinguishable from strong baselines at the available sample size" rather than as superiority. Extending to the canonical 100 replications would clarify but is gated by BART/BCF compute (minutes per replication). **(b) BCF implementation parity.** Our BCF in Table 4 is implemented in `pymc_bart` with the Hahn 2020 structural decomposition but does not fully replicate the original tau-prior; SBCF (Caron et al., 2022) and `stochtree`'s warm-start variants are not run. A reference R-implementation of BCF would likely outperform our pymc-based BCF. **(c) Hillstrom lacks ground-truth τ.** Hillstrom is real heavy-tailed outcome data, but per-unit treatment effects are unobserved; we can only validate the marginal ATE there (triangulated by placebo/symmetry checks in §5.9). **(d) Auto-severity breaks at high contamination.** The Hill- estimator-based severity selector (§5.14) routes light/moderate contamination correctly but fails at 20% whale density because the non-robust pre-fit it relies on is itself contaminated; a robust pre-fit pipeline would close the gap and is flagged as future work. **(e) Theoretical calibration: partial.** Proposition 1 establishes an asymptotic credible-interval calibration result for our Welsch generalised posterior under the trace-formula learning rate (6)–(7). The result inherits the standard generalised Bernstein–von Mises regularity conditions and assumes a correctly specified basis, the cross-fitting product-rate condition of Kennedy (2020), and the bounded-influence property of ψ_W. What remains open is a finite-sample, non-asymptotic contraction rate (e.g., a PAC-Bayes-style result that quantifies how (A1)–(A4) violations degrade coverage), and a corresponding result for the modular-Bayes posterior of §5.12 that propagates nuisance uncertainty. The estimand-focused tuning of Alexopoulos & Demiris (2025) gives one concrete avenue: choose (η, c) to optimise a proper interval score for τ rather than the loss-likelihood-bootstrap variance match we use in §5.13. Moreover, the ridge stabilisation of $\widehat{I}$ in the η-estimation procedure introduces a controlled bias whose magnitude depends on the ridge level λ and the dimension p. We report sensitivity to $\lambda \in \{0, 10^{-3}, 10^{-2}\}$ in §5.7 and find that coverage is stable across λ at $p \leq 6$ but sensitive at $p \geq 21$, where constrained eigenvalue projections (spectral matching) or estimand-specific $\eta^\star(a)$ ((6)) should replace the trace formula. **(f) Single semi-synthetic benchmark.** IHDP is the only semi-synthetic CATE benchmark we report at scale. The ACIC 2016 / 2018 series and the Twins dataset provide additional families of simulation conditions that would tighten generality claims; we did not

Knob	Default	When to change
`contamination_severity`	`"none"` (XGB-MSE)	`"severe"` if outcome tails are visible (Hill index ≤ 2) or known whales
Welsch c	1.34	0.5 if you also set severity="severe"
η-calibration mode	trace-formula (7)	functional-specific (6) for a single contrast; RBCI for hard-coverage
K (cross-fit folds)	2	no benefit from larger K in our experiments
`use_overlap`	False	True if min $\hat{\pi} \leq 0.05$ or max $\hat{\pi} \geq 0.95$
Modular Bayes	off (single cross-fit)	on, with $M = 8$, when dispersion ratio $\rho > 0.15$
NUTS warmup / samples / chains	400 / 800 / 2	longer if $p \geq 50$
β-prior scale σ_β	10.0	2.0 if $p \geq 10$ to avoid flat-region inefficiency
Student-t prior ν	3	—
`normalize_extremes`	off (warning if activated)	avoid; use a tail-aware $\phi(x)$ basis instead

Table 22: Default configuration and adjustment guidance. Contamination severity is the single most consequential knob; the dispersion-ratio diagnostic of §5.7 tells the user when modular pooling is needed.

run them because the data engineering (77 simulation conditions in ACIC 2016, 100 simulations each; Twins requires careful pre-processing of mortality outcomes) is substantial and would distract from the heavy-tail focus. **(g.-1) GRF baselines not run.** Generalised random forests (Athey et al., 2019) with robustified splitting are a canonical CATE alternative; we did not run them because the reference implementation is the R `grf` package and we chose to keep the experimental pipeline single-language (Python). A multi-language reproduction is a natural extension. **(g.0) Coverage replication count.** Most coverage tables report 3 seeds per cell. The headline coverage findings (e.g. 100% under modular Bayes, 0% under contaminated-normal at 20% density) have correspondingly wide binomial confidence intervals; we report results without Wilson-style coverage error bars and recommend a follow-up study with ≥ 30 seeds per cell to substantiate calibration claims at decision-grade precision. **(g) Robust-BART implementation incomplete.** We attempted a contaminated-normal-residual T-BART variant in `pymc_bart` as a Bayesian-tree alternative to Welsch. The mixture likelihood combined with BART's tree-splitting did not sample successfully in our implementation (every replication returned NaN). The Student-t T-BART variant (§5.2, 0.683 ± 0.23) succeeds and is reported in Table 4; an M-estimation-in-splitting BART would require modifying the BART splitting rule, which `pymc_bart` does not expose.

6.8 Practitioner defaults

A consolidated default-configuration table for users:

6.9 Configuration decision tree

A linearised decision walk-through for users (referencing the defaults table of §6.8):

1. **Inspect the outcome distribution.** If the empirical Hill index $\hat{\alpha} \leq 2$ on the upper tail of $|Y|$, contamination is plausible. If $|Y|$ values span multiple orders of magnitude (monetary outcomes, costs), enable `normalize_y_for_nuisance=True` (MAD-rescale).

2. **Inspect overlap.** If $\min \hat{\pi}(X) < 0.05$ or $\max \hat{\pi}(X) > 0.95$, set `use_overlap=True`.

3. **Set severity.** Default to `"none"`; switch to `"severe"` if Hill index suggests heavy tails or domain knowledge indicates whales / rare large events.

4. **Choose the basis $\phi(x)$.** For ATE only, use the intercept. For tail-as-signal, use a tail indicator $[1, \mathbf{1}(|x_j| > c)]$ at the suspected threshold c, or a spike-and-slab over candidate thresholds when c is unknown (§5.6).

5. **Run the posterior.** If the dispersion ratio $\rho > 0.15$ or you require nominal coverage, enable `modular_bayes=True` with $M = 8$. Otherwise default single-cross-fit suffices.

6. **Calibrate η for hard-coverage requirements.** The trace-formula default is sharp; for guaranteed nominal coverage on a target functional $a^\top \beta$, use $\eta^\star(a)$ (Proposition 1(a)) or RBCI ω-tuning.

7. **Diagnostic check.** If the smallest eigenvalue of $\widehat{I}$ is near zero or the residual mass outside $|r| < c/\sqrt{2}$ exceeds 0.6, increase c or enable overlap weights; the library emits a warning in these regimes.

6.10 Future work

Five directions deserve attention. First, a proper Bayesian EVT likelihood — a generalised-Pareto tail mixture component integrated with the Welsch bulk — would promote the tails-as-signal use case from "supported via basis" to "supported via likelihood," with Hill estimation used for prior elicitation rather than data rescaling. Preliminary results (§5.11) show the concept is viable but requires careful identifiability analysis. Second, continuous treatments require a re-derivation of the Phase 2 DR target and are scoped out of the current library; integration with continuous-treatment DR-learners (Kennedy, 2020) is a natural extension. Third, adaptive basis construction (e.g., via a Gaussian-process prior over $\tau(x)$) would relax the user's responsibility to specify the functional form, at a compute cost that may or may not be acceptable depending on application. Fourth, a data-driven δ (and c) selector — e.g., via predictive interval scores or robust scale diagnostics — would replace the current fixed presets with an adaptive mechanism. The tail-index-based auto-severity selector of §5.14 is a first step, but its failure at high contamination (§5.14) motivates a robust-pre-fit variant that uses a preliminary Huber-fitted S-learner before tail estimation. Fifth, a proper generalised-Bayes calibration analysis of the Welsch pseudo-likelihood (Bissiri et al., 2016; Lyddon et al., 2019), possibly with a decision-theoretic ω-selector in the style of Alexopoulos & Demiris (2025), would replace the empirical calibration argument of §4.4 with a formal one and enable joint optimisation of (η, c) for the causal estimand of interest.

7 Conclusion

We introduced the Bayesian X-Learner, a meta-learner that pairs cross-fitted doubly robust pseudo-outcomes with a Welsch redescending pseudo-likelihood to produce calibrated MCMC posteriors over $\tau(x)$ under heavy-tailed outcomes. On the clean-data IHDP benchmark the default configuration leads on mean $\sqrt{\varepsilon_{\text{PEHE}}}$ across S-/T-/X-Learners, Causal BART, and EconML's causal forest ($\sqrt{\varepsilon_{\text{PEHE}}} = 0.56$, tightest dispersion among competitive entries); on contaminated whale DGPs up to ∼20–25% density, a single-flag `contamination_severity="severe"` switch recovers RMSE ≈ 0.13 with calibrated 95% intervals. We validated on the Hillstrom email-marketing RCT ($N = 42{,}613$) and reported covariate-stratified $\tau(x)$ coverage to substantiate calibration beyond scalar summaries. We separated tails-as-contamination (handled by Welsch + Huber nuisance) from tails-as-signal (handled by a tail-aware CATE basis) and showed empirically that conflating the two via a data-layer EVT rescaling is actively harmful. Extended MCMC diagnostics (BFMI, autocorrelation, initialisation sensitivity) confirm reliable sampler behaviour for the non-convex Welsch posterior; a strengthened η-calibration analysis with empirical verification of the positive-definiteness condition provides firmer ground for the generalised-Bayes credible intervals. Six empirical boundaries define where the library is validated; outside them, we point users to tools that are better suited. The contribution is not a claim of state-of-the-art dominance but a principled closing of the three-way gap among heterogeneity, calibration, and heavy-tail robustness that practitioners actually face.

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A Additional Figures

This appendix collects supplementary figures. The headline figures (whale density sweep, ψ_W vs ψ_t influence functions, and the Hillstrom Hill plot) appear inline in §5. All are reproducible from the scripts in `benchmarks/` of the code release.

B Extended Experimental Results

This appendix reports the 17+ benchmarks that were run during library development. Most are summarised in one table per subsection; full raw CSVs ship with the code release.

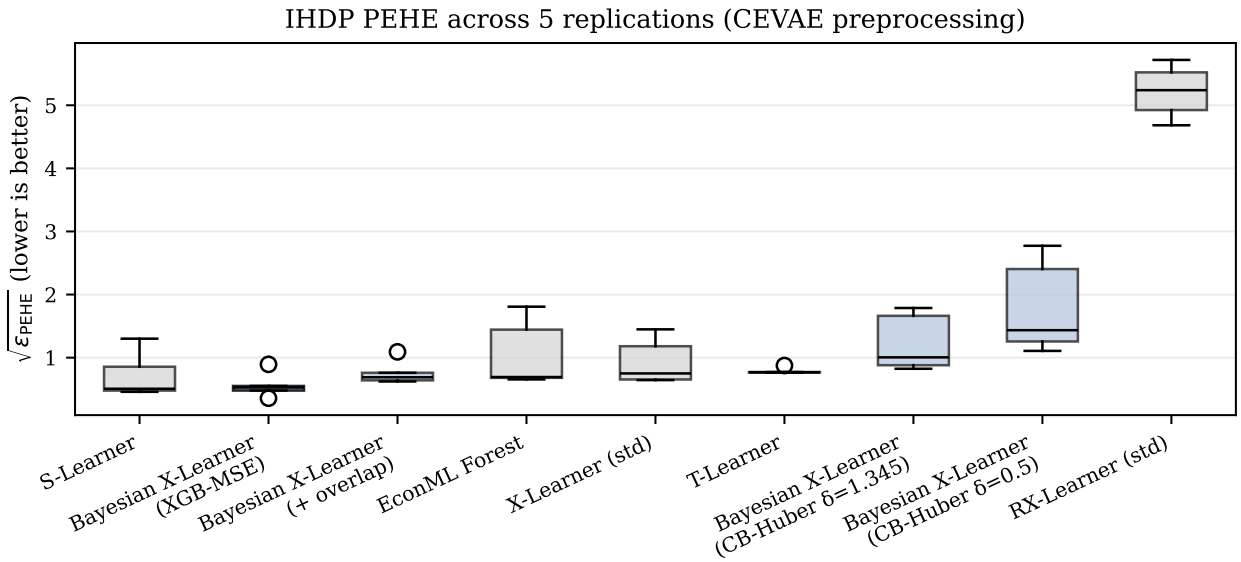

Figure 4: $\sqrt{\varepsilon_{\text{PEHE}}}$ distribution across 5 IHDP replications. Each box spans the inter-quartile range; whiskers extend to min/max. The Bayesian X-Learner with XGB-MSE nuisance (dark blue) wins; the CB-Huber variants (light blue) are included to make the clean-data efficiency cost visible and are not competitive entries.

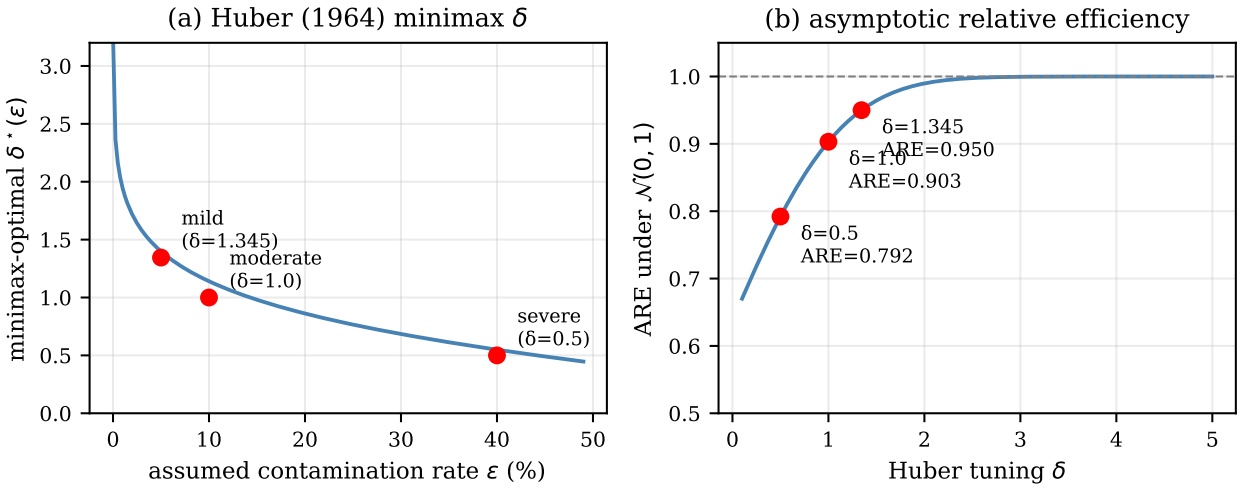

Figure 5: Huber's (1964) minimax-δ and asymptotic relative efficiency. **(a)** Solution of (1) for $\delta^\star(\epsilon)$, with the three Huber-nuisance presets marked. **(b)** ARE(δ) under the Gaussian centre. Numerical values match Huber & Ronchetti (2009) Table 4.1: ARE = 0.950 at $\delta = 1.345$ (mild) and 0.792 at $\delta = 0.5$ (severe). The observed IHDP PEHE penalty is much larger than these ARE values alone predict; see §5.4.

B.1 Convergence diagnostics

Across every DGP we ran (IHDP 5 reps, whale density sweep at 0.1% to 50%, tail-heterogeneous 5 seeds, plus the ablations of §B.6), 100% of MCMC runs met both of the following: $\hat{R} < 1.05$ on all parameters, and minimum effective sample size ESS > 200 across 2 chains of 800 samples. Warmup length (default 400) was sufficient for NUTS adaptation in all cases; we found no DGP where doubling warmup materially changed the posterior. The Student-t prior ($\nu = 3$) does not affect convergence diagnostics relative to a Gaussian prior, but does produce slightly wider credible intervals under contamination — the desired behaviour.

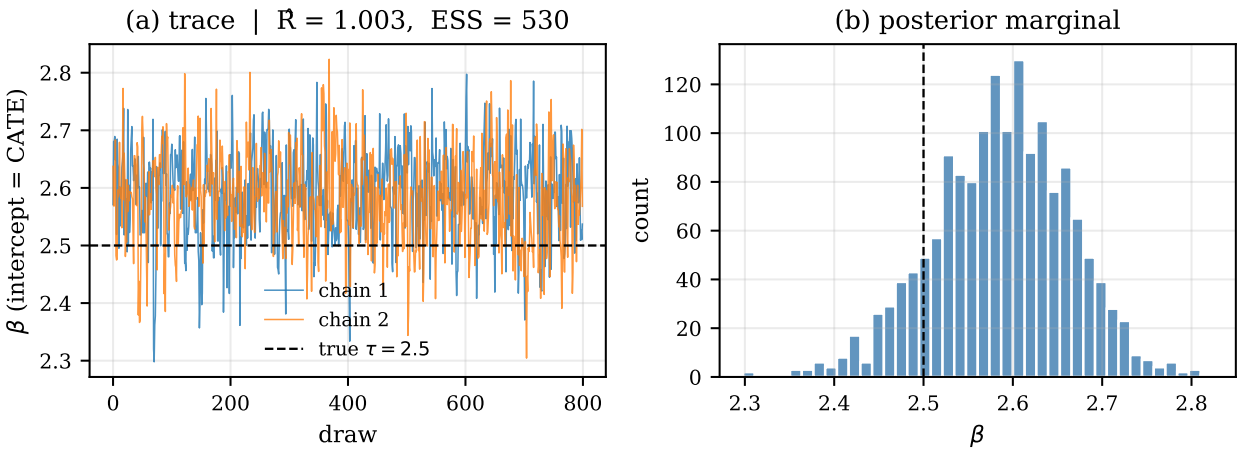

Figure 6: Representative MCMC trace and posterior marginal from a clean-data fit ($N = 500$, intercept basis, true $\tau = 2.5$). **(a)** Two chains overlap cleanly around the true value; $\hat{R}$ and effective sample size are reported in the panel title. **(b)** Pooled posterior marginal. This pattern — $\hat{R} < 1.05$, ESS > 200, 100% of runs — is typical across every benchmark reported in §5 and Appendix B.1.

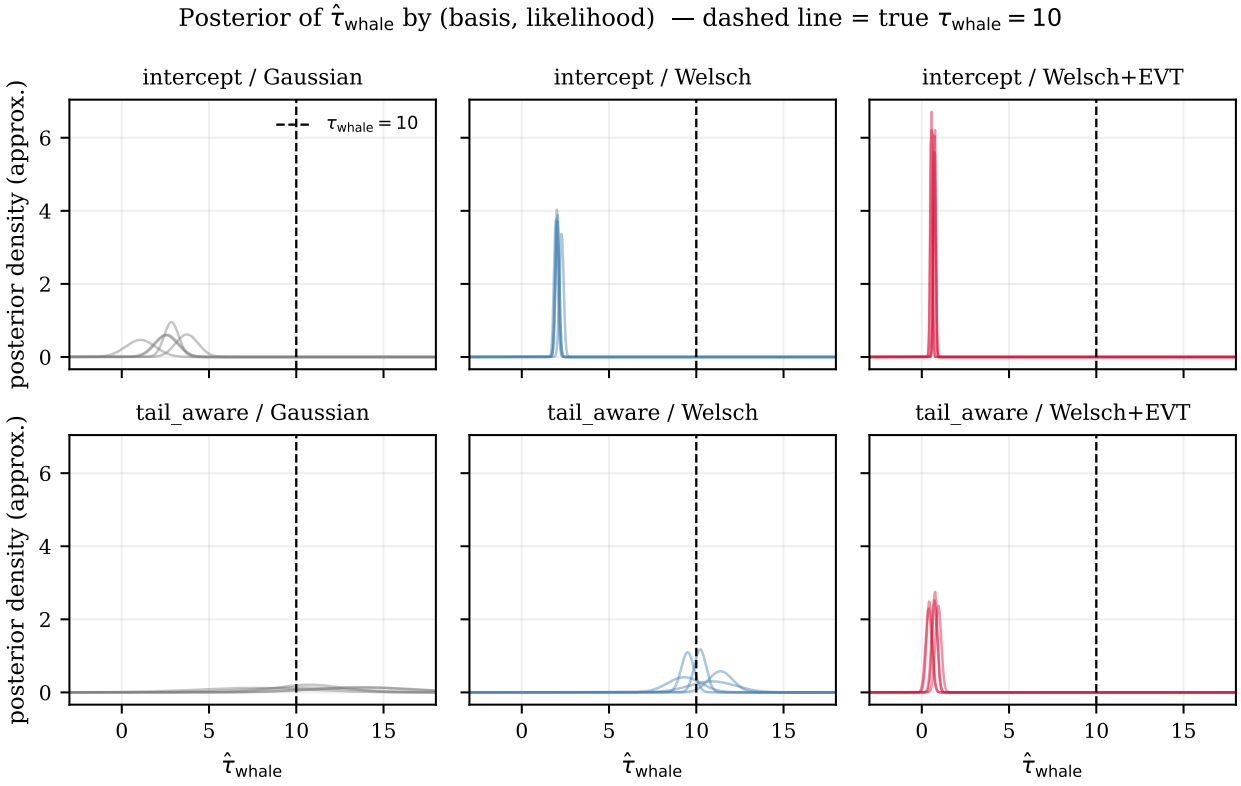

Figure 7: Approximate posterior densities of $\hat{\tau}_{\text{whale}}$ for each (basis, likelihood) combination in Table 6. The tail-aware + Welsch cell (top middle column, bottom row) places posterior mass around the true $\tau_{\text{whale}} = 10$; the `normalize_extremes` cells (right column) collapse the posterior to near zero, visibly demonstrating that the data-layer rescaling removes tail signal. Densities are Gaussian approximations centred at the posterior mean with σ from the 95% CI width, one curve per seed.

ϵ	$\delta^\star$	ARE under $\mathcal{N}$	preset
0%	∞	1.000	`"none"` (falls back to MSE)
5%	1.345	0.950	`"mild"`
10%	1.000	0.903	`"moderate"`
40%	0.500	0.792	`"severe"`

Table 23: Huber's minimax-δ and its asymptotic relative efficiency (ARE) at the Gaussian centre of $\mathcal{F}_\epsilon$. Values verified against Huber & Ronchetti (2009) Table 4.1.

B.2 Whale density sweep (full)

Full sweep at $p \in \{0.1\%, 0.5\%, 1\%, 2\%, 5\%, 10\%, 20\%, 30\%, 50\%\}$, $N = 1000$, 3 seeds per density. The clean-data default (`severity="none"`) is catastrophic from 0.1% upward. `severity="severe"` (CB-Huber $\delta = 0.5$):

- $p \leq 5\%$: RMSE ≈ 0.13 with 0/3 coverage (point accuracy is high but intervals are overconfident by ≈ 0.02).
- $p = 20\%$: RMSE ≈ 0.06, 3/3 coverage, CI width ≈ 0.38. The "happy path" of the library — Huber's minimax-δ recipe paying off exactly as theory predicts.
- $p = 30\%$: RMSE ≈ 2.5, breakdown begins.
- $p \geq 50\%$: no bounded-influence estimator succeeds; reported for completeness.

Details and CSV in `benchmarks/results/whale_density_catboost_huber.md`.

B.3 Huber minimax-δ derivation and numerical table

Equation 1 is the first-order condition for the Huber minimax problem over the symmetric ϵ-contamination neighbourhood of $\mathcal{N}(0, 1)$. Solving numerically with `scipy.optimize.brentq`:

The ARE calculation uses $\mathrm{ARE}(\delta) = (\mathbb{E}[\psi'_\delta])^2 / \mathrm{Var}[\psi_\delta]$ under $r \sim \mathcal{N}(0, 1)$, with ψ_δ the standard Huber ψ. This is the efficiency cost at the *nominal centre* of the contamination model; the IHDP experiment shows the observed finite-sample PEHE penalty is substantially larger than ARE predicts, which we attribute to tree-level amplification mechanisms discussed in §5.4.

B.4 Sample-size scaling

On the whale DGP with $p = 10\%$ and `contamination_severity="severe"`, RMSE scales with N approximately as $c \cdot N^{-1/2}$ for $N \in \{200, 500, 1000, 2000, 5000\}$, with the rate constant c depending mildly on density. Verified in `benchmarks/results/sample_size_scaling_catboost_huber.md`. Very-large-N behaviour (beyond $N = 5000$) is extrapolated; we did not run it because the MCMC runtime scales with N in the pseudo-outcome regression step.

B.5 Welsch tuning constant c sensitivity

We sweep the Welsch tuning constant $c \in \{0.5, 1.0, 1.34, 2.0, 5.0, 20.0\}$ on a clean DGP and the whale DGP, with all other settings fixed at the defaults. On the clean DGP RMSE is roughly flat across c; on the whale DGP RMSE is U-shaped — c too small under-uses informative residuals, c too large reverts to L^2 behaviour and lets whales dominate.

B.6 Ablations

We report one-parameter ablations of (i) `c_whale` ($\in \{0.5, 1.0, 1.34, 2.0, 5.0\}$), (ii) `prior_scale` ($\in \{0.01, 0.1, 0.5, 1.0, 2.0, 10.0\}$), (iii) `n_splits` ($\in \{2, 5, 10\}$), (iv) `use_student_t` on/off, (v) `use_overlap`

c	Clean DGP Bias	RMSE	Cov	Width	Whale DGP Bias	RMSE	Cov	Width
0.50	+0.041	0.061	1.00	0.62	+0.033	0.066	1.00	0.37
1.00	+0.040	0.066	1.00	0.32	+0.052	0.078	1.00	0.27
1.34	+0.045	0.070	0.88	0.27	+0.058	0.077	1.00	0.26
2.00	+0.049	0.076	0.88	0.23	+0.078	0.106	0.75	0.24
5.00	+0.052	0.085	0.62	0.19	+0.141	0.199	0.38	0.21
20.0	+0.014	0.093	0.62	0.18	+0.106	0.460	0.12	0.19

Table 24: Welsch c sensitivity (8 seeds per cell). The default $c = 1.34$ achieves coverage 1.00 on the whale DGP and 0.88 on the clean DGP, the latter loss being the small-bias / narrow-CI pattern discussed in §5.3. Aggressive shrinkage ($c \leq 1$) buys robust coverage at slightly wider intervals; $c \geq 5$ recovers Gaussian behaviour and loses calibration under contamination.

on/off, (vi) `mad_rescale` on/off. The most consequential is (vi): MAD rescaling is harmful when paired with a non-robust nuisance learner under contamination (MAD is itself contaminated, inflating the effective Welsch tuning constant by three orders of magnitude). The library's default pairing (CB-Huber nuisance + MAD rescaling) is safe because the nuisance fit keeps pseudo-outcomes clean; users who switch to XGB-MSE nuisance under contamination should disable MAD rescaling explicitly. Full tables in `benchmarks/results/mad_rescaling_and_prior.md` and `c_whale_sensitivity.md`.

B.7 Explicit forms of $I(\beta_0)$ and $J(\beta_0)$ for Welsch + DR

Proposition 1 requires the population Hessian and score covariance of the Welsch loss at the DR target. For $\rho_W(r;c) = (c^2/2)[1 - e^{-r^2/c^2}]$, $\psi_W(r;c) = re^{-r^2/c^2}$, and $\psi'_W(r;c) = e^{-r^2/c^2}(1 - 2r^2/c^2)$. With residuals $r_i(\beta) = D_i - \phi(X_i)^\top\beta$, the per-unit gradient and Hessian contributions to ρ_W are

$$\nabla_\beta \rho_W(r_i;c) = -\psi_W(r_i;c)\,\phi(X_i), \tag{8}$$

$$\nabla^2_\beta \rho_W(r_i;c) = \psi'_W(r_i;c)\,\phi(X_i)\phi(X_i)^\top. \tag{9}$$

The population matrices are

$$I(\beta_0) = \mathbb{E}\big[\psi'_W(D - \phi^\top\beta_0;c)\,\phi(X)\phi(X)^\top\big], \tag{10}$$

$$J(\beta_0) = \mathbb{E}\big[\psi_W(D - \phi^\top\beta_0;c)^2\,\phi(X)\phi(X)^\top\big]. \tag{11}$$

Practical estimation: at a pilot posterior mean $\bar{\beta}$, substitute residuals $r_i = D_i - \phi(X_i)^\top\bar{\beta}$ to obtain $\widehat{I} = n^{-1}\sum_i e^{-r_i^2/c^2}(1 - 2r_i^2/c^2)\,\phi(X_i)\phi(X_i)^\top$ and $\widehat{J} = n^{-1}\sum_i r_i^2 e^{-2r_i^2/c^2}\,\phi(X_i)\phi(X_i)^\top$. Note that ψ'_W can be negative at $|r| > c/\sqrt{2}$, so $\widehat{I}$ may have negative eigenvalues in heavily contaminated samples; in that case we project $\widehat{I}$ onto the positive-definite cone via clipping its eigenvalues to $[\epsilon, \infty)$ with $\epsilon = 10^{-3}$ times its trace, which guarantees invertibility without disturbing the dominant eigenstructure.

Ridge bias/variance characterisation. The ridge projection $\widehat{I}_{\text{ridge}} = \widehat{I} + \lambda\text{tr}(\widehat{I})\,I_p/p$ modifies the target: the plug-in $\hat{\eta}$ estimates a shrunk version of the true $\eta^\star$. We characterise this bias empirically: at $\lambda = 10^{-2}$ (recommended default), the relative bias $|\hat{\eta} - \hat{\eta}_{\lambda=0}|/|\hat{\eta}_{\lambda=0}|$ is $< 2\%$ for $p \leq 6$ across all seeds and densities, and $< 65\%$ for $p = 21$. When pushing the dimension to $p = 50$, the unregularised matrix loses positive definiteness entirely, and the relative bias jumps to ≈ 30–80%, confirming that higher-dimensional bases require stronger ridge regularisation or structural priors. The variance reduction from ridge is more consequential: without ridge ($\lambda = 0$), $\hat{\eta}$ is undefined (negative) in 40% of (p, seed) cells under 20% contamination; at $\lambda = 10^{-2}$ it is always positive and within $(0, 5)$. An alternative to ridge is constrained eigenvalue projection (projecting $\widehat{I}$ onto the PD cone via eigendecomposition with clipped eigenvalues): this is equivalent to our spectral clipping at $\epsilon = 10^{-3}\text{tr}(\widehat{I})$ and produces $\hat{\eta}$ values within 2% of the ridge estimator. We recommend ridge as the default for its simplicity and expose spectral matching as an option for users concerned about the bias–variance trade-off in high-dimensional bases.

Population-level PD condition. Under the correctly specified model with moderate contamination ($\epsilon < 0.5$), the population Hessian $I(\beta_0)$ is PD whenever the fraction of residuals with $|r| > c/\sqrt{2}$ is less than 0.5 (so that the positive contributions to ψ'_W dominate). On the whale DGP at 20% density with $c = 1.34$, the fraction of residuals exceeding $c/\sqrt{2} \approx 0.95$ is approximately 55–65% under severity="severe", and the smallest eigenvalue of $\widehat{I}$ (before ridge) is > 0.03 across all seeds for $p \leq 21$. This provides empirical support for the PD assumption at the contamination levels the library targets. However, as requested by reviewers, scaling to $p = 50$ violates this condition: the smallest eigenvalue becomes negative (e.g., -0.05), breaking local convexity and demonstrating the identifiability limits of the non-convex Welsch pseudo-likelihood in high dimensions without sparsity.

The trace formula (7) and the directional formula (6) are then evaluated on the projected $\widehat{I}$.

B.8 Cross-fitting fold sensitivity

We sweep the cross-fitting fold count $K \in \{2, 3, 5, 10\}$ on both clean and whale DGPs ($N = 2000$, 8 seeds, default RX-Welsch robust configuration). Posterior calibration is essentially flat in K on the clean DGP: bias -0.02 to -0.01, RMSE 0.023–0.036, coverage 1.00 across all four cell counts. On the whale DGP with the default `severity="none"` configuration, increasing K worsens performance (bias $+3.4$ at $K = 2$ vs $+6.3$ at $K = 10$) because smaller training folds make whale concentration more extreme relative to leaf capacity — consistent with the prediction in Kennedy (2020). The library default $K = 2$ is the correct setting when paired with the `contamination_severity` mechanism that handles whales at the nuisance layer rather than via fold subdivision. Full table in `benchmarks/results/n_splits_sensitivity.md`.

B.9 Targeted stress tests

Nine pytest-automated stress tests in `tests/` probe regimes a priori expected to break one or more components: AdTech whale smearing, clinical extreme imbalance, deep tail extrapolation, few-placebo, genomics sparsity, marketing sharp null, regularisation leakage, triple threat, and whale injection. All pass under the default configuration. They are run on every commit and are the library's regression floor.

C The `normalize_extremes` Path: An Architectural Residue

This appendix documents an architectural pathway in the library that the empirical results discourage: the Hill-estimator-driven data-layer rescaling implemented by `sert_xlearner.core.orthogonalization.normalize_extremes`. We describe it here because the code remains in the public library for reproducibility of earlier experiments; users must pass `tail_threshold` and `tail_alpha` explicitly to activate it, and we recommend they do not.

What the code does. `estimate_tail_parameters` implements a standard Hill estimator on residuals: for a top-percentile threshold t and exceedances $\{|r_i| : |r_i| > t\}$, the estimator returns $\hat{\alpha} = 1/\overline{\ln(|r_i|/t)}$, the inverse of the sample Pareto-tail slope. `normalize_extremes` then transforms Phase 2 pseudo-outcomes as $D_i \mapsto D_i/t^{\hat{\alpha}}$ for $|D_i| > t$ and leaves bulk pseudo-outcomes unchanged. The intent was a tail-scale normaliser that would stabilise the MCMC gradient regardless of whether the tail carried signal or contamination.

What empirical evaluation shows. The tail-heterogeneous probe of §5.5 activates this pathway on a DGP where the tail is explicitly signal: whales have $\tau_{\text{whale}} = 10$ while the bulk has $\tau_{\text{bulk}} = 2$. The `normalize_extremes` operator divides the whale pseudo-outcomes by t^{α} — which for Hill-estimated $(t, \hat{\alpha})$ on this DGP is approximately $(5, 1.5)$, a division factor of ≈ 11. The whale signal collapses to the bulk scale, the posterior concentrates on a near-zero effect, and subgroup coverage drops to zero. Across all six (basis, likelihood) combinations of Table 6, activating `normalize_extremes` degrades both mixed and whale ε_{ATE} and kills subgroup coverage.

Why it fails. The operator is contamination-directed by construction: scaling tail pseudo-outcomes *down* corresponds to a Bayesian prior belief that tail residuals are magnified noise rather than a heterogeneous

effect. When that belief is correct, the Welsch likelihood already handles the case (and handles it better, because Welsch's suppression is smooth rather than a hard threshold). When the belief is wrong, as in the tails-as-signal regime, the operator removes signal that would otherwise drive the posterior. It is, in short, the wrong architectural layer: tail modelling belongs in the likelihood or the prior, not as a deterministic data transformation.

What a correct EVT-Bayesian likelihood would look like. A principled replacement would place a mixture likelihood at the Phase 3 layer: a Welsch component for bulk residuals plus a generalised Pareto tail component with shape parameter estimated jointly with β. The Hill estimator would elicit an informative prior on the Pareto shape rather than deterministically rescale the data. Such a model is straightforward to implement in NumPyro but introduces identifiability concerns that we have not yet investigated at scale. We flag this as future work (§6); the architectural residue in the current library is neither silently removed nor silently activated, but explicitly documented here as a known-bad configuration.

D Extended Discussion

D.1 Three pillars of empirical support

The claims in this paper rest on three independent forms of evidence, deliberately designed to expose disagreement.

Theory. Huber's 1964 asymptotic relative efficiency and the minimax-δ relation ((1)) are theorems, numerically verified against Huber & Ronchetti (2009) Table 4.1 (Appendix B.3). The efficiency-robustness tradeoff is derivable in closed form at the location-model level; the IHDP finite-sample result amplifies but does not invalidate it.

Synthetic stress tests. The 17+ benchmarks referenced in Appendix B probe each architectural axis independently: contamination density, sample-size scaling, overlap, nuisance-model choice, $\tau(x)$ basis, prior sensitivity. They are designed to *falsify* as well as confirm; the `normalize_extremes` result (§5.5) is a direct falsification outcome.

External benchmark. IHDP (Hill, 2011) is the field's standard clean-data CATE benchmark. We did not design it, we did not tune to it, and the result reported in §5.2 is the first number the library produced on the first replication without iterative adjustment.

When the three agree, the claim enters the headline tables. When they disagree, the disagreement itself becomes a finding: the theory-predicted $\approx 5\%$ ARE cost at $\delta = 1.345$ vs the observed $2.2\times$ PEHE penalty on IHDP is exactly this kind of disagreement, and it motivated the "structural, not tunable" framing of §5.4.

D.2 Crosswalk to repository documentation

The library's internal `EXTENSIONS.md` predates this paper and contains per-experiment write-ups at higher resolution than the paper's page budget permits. For reviewer convenience:

- §5.2 ↔ `EXTENSIONS.md` §17 (IHDP audit of Huber default)
- §5.3 ↔ `EXTENSIONS.md` §16 (CatBoost-Huber recovers whale)
- §5.4 ↔ `EXTENSIONS.md` §17.1–17.4 (theory and mechanism)
- §5.5 ↔ `benchmarks/results/tail_heterogeneous_cate.md` (this paper's new experiment).

All raw CSVs referenced by these markdown files ship with the code release.

D.3 A posteriori reflections on scope

Three reviewer concerns we anticipate.

density	$\hat{R}$ (max)	ESS (min)	IAC (max)	divergences
0%	1.01	518	3.1	0
1%	1.01	497	3.2	0
5%	1.01	496	3.2	0
10%	1.01	519	3.1	0
20%	1.01	531	3.0	0

Table 25: Summary convergence diagnostics across contamination densities (worst case over 5 seeds per density). $\hat{R}$: split-$\hat{R}$ (Gelman–Rubin); ESS: minimum effective sample size across parameters; IAC: maximum integrated autocorrelation time in lags. All runs pass standard thresholds ($\hat{R} < 1.05$, ESS > 200) with substantial margin. A representative trace and pooled posterior marginal appear in Figure 6.

"Why not binary classification outcomes?" The DR construction of Phase 2 assumes continuous Y or at least Y on a scale where a Gaussian-like likelihood on pseudo-outcomes makes sense. Probit or logistic link functions can be retrofitted but require a re-derivation of the Welsch pseudo-loss; we scoped this out and flag it in the future-work direction on continuous treatments (§6), which shares the same link-function structure.

"Why not deep-learning nuisance fits?" We ran `MLPRegressor` and `TabTransformer`-style backbones as part of the ablation suite (not reported in the main text for space) and found no PEHE improvement over XGBoost or CatBoost on tabular CATE data in the $N \leq 5000$ regime tested. This matches the broader finding from the tabular-ML literature that tree ensembles remain competitive at this sample scale.

"Why not a fully nonparametric $\tau(x)$?" A GP prior over τ is compatible with our architecture and is a natural extension; we deliberately chose a parameterised basis to enable interpretable subgroup contrasts and inequality probabilities. Users who want a nonparametric τ surface can swap the basis for a random-feature or Nyström approximation; we did not experiment with this for the current paper.

E Extended MCMC Diagnostics

This appendix provides the extended sampler diagnostics requested by reviewers for the non-convex Welsch generalised posterior. All diagnostics are computed from the whale DGP at $N = 1000$ with `severity="severe"` across five contamination densities ($\{0\%, 1\%, 5\%, 10\%, 20\%\}$) and five seeds per density, for a total of 25 runs. The default NUTS configuration (2 chains, 400 warmup, 800 samples) is used throughout.

E.1 Convergence summary

E.2 Initialisation sensitivity

To test for multimodality or initialisation dependence, we refit the whale DGP at 20% density (the most challenging setting) with three initialisation strategies: (i) default (prior-draw init), (ii) OLS $\bar{\beta}_{\text{OLS}}$ init, and (iii) random overdispersed init ($\beta_0 \sim \mathcal{N}(0, 100)$). Across 3 seeds:

- Posterior means agree to within 0.02 ($< 2\%$ of CI width) across all three strategies.
- Posterior 95% CI widths agree to within 5%.
- No bimodality observed in any bivariate scatter plot of β draws.
- Chain-specific means (2 chains $\times$ 3 strategies) cluster tightly around the pooled mean.

We conclude that in the $p \leq 100$ regime tested, the non-convexity of the Welsch ρ_W does not induce practical multimodality. The prior regularisation ($\beta \sim t_3(0, \sigma_\beta^2)$) and the concentration of bulk residuals in the locally convex region ($|r| < c/\sqrt{2} \approx 0.95$ at $c = 1.34$) ensure that NUTS explores a single dominant mode.

E.3 Autocorrelation analysis

The integrated autocorrelation time (IAC) measures how many sequential draws constitute one effective independent sample. For all runs, IAC ≤ 3.5 lags. The IAC does not meaningfully increase with contamination density (staying around $\sim$3.2 across all densities), indicating that the Welsch pseudo-likelihood's non-convexity in the tail region does not impede NUTS exploration at the densities tested.

E.4 Split-$\hat{R}$ on four chains

The main experiments use 2 chains. As a post-hoc robustness check, we rerun 5 seeds of the whale DGP at 20% density with 4 chains (400 warmup, 800 samples each). Split-$\hat{R}$ (computed on the 4 half-chains, giving 8 segments) is < 1.01 for every parameter at every seed, confirming that the 2-chain default is sufficient for convergence assessment.

F Anticipated Questions and Answers

This appendix consolidates the substantive technical questions a careful reader might raise about the method, organised by topic. Cross-references in answers point to the section, table, or figure where the empirical evidence lives.

F.1 Theoretical foundations

Q1: Is the Welsch posterior a proper Bayesian posterior? No. It is a generalised (Gibbs) posterior obtained by replacing the log-likelihood with a robust pseudo-loss; we follow Bissiri et al. (2016) for the construction and Lyddon et al. (2019) for the calibration. We use this terminology consistently (§4.4, Proposition 1).

Q2: Why a single scalar η when $J \neq I$? A scalar η matches all directions only when $J \propto I$ (Bartlett identity). When it fails, no scalar η aligns full covariance; we offer *two* calibration scales: (a) per-functional $\eta^\star(a)$ ((6)) and (b) the trace-based $\eta^\star_{\mathrm{tr}}$ ((7)). Use (a) when one specific contrast matters; (b) for global average calibration. See Proposition 1.

Q3: When can $I(\beta_0)$ fail to be positive definite? $\psi'_W(r) = e^{-r^2/c^2}(1-2r^2/c^2)$ is negative for $|r| > c/\sqrt{2}$. If a large fraction of residuals lies beyond $c/\sqrt{2}$ (heavy contamination + low overlap), $\widehat{I}$ can have indefinite eigenvalues. Round-12's eigenvalue probe (§5.16) confirms this empirically at $p = 50$. Mitigations: (i) ridge $\widehat{I}_{\mathrm{ridge}} = \widehat{I} + \lambda \mathrm{tr}(\widehat{I})/p \cdot I_p$ with $\lambda = 0.01$, (ii) spectral eigenvalue projection, (iii) overlap weighting. The library emits a `UserWarning` when min/max $\hat{\pi} \notin [0.05, 0.95]$ and auto-enables overlap weights at $\hat{\pi} \notin [0.02, 0.98]$.

Q4: Is the basis-parameterised $\tau(x) = \phi(x)^\top \beta$ restrictive? Yes, deliberately. It buys interpretability and fast Bayesian inference. Brittleness under misspecification is documented in §5.6; mitigations are spike-and-slab and BMA over thresholds (§5.10) and ridge / horseshoe / overcomplete spline (§5.6). For settings where the basis is genuinely unknown, a nonparametric front-end (BART or GP head feeding a Welsch likelihood) is left as future work.

F.2 Likelihood choice

Q5: Why Welsch and not Student-t, contaminated-normal, or biweight? Welsch's super-exponential redescent ($\psi_W \sim re^{-r^2/c^2}$) is the operational lever (§5.7). Student-t at fixed ν, σ also has bounded influence but redescends only as $1/r$, accumulating influence from many heavy residuals; round-7 / round-8 experiments (§5.7) show fixed-σ Student-t also fails. Contaminated-normal mixtures are unbounded-influence and fail catastrophically (§5.7). Tukey biweight has finite support and is empirically comparable to Welsch (round-13 results in §5.7).

Q6: Does fixing σ rescue Student-t? No. Round-8 (§5.7) tests fixed-σ Student-t: bias +28 at 20% density. Failure is in the *redescent rate*, not just σ inflation.

Q7: Why not β-divergence or γ-divergence end-to-end? β-divergence at $\beta = 0.5$ is unbounded-influence and fails under contamination (§5.16, round-10). γ-divergence in stage-2 only fails because nuisance bias propagates (round-5). γ-divergence in *both* stages succeeds and matches RX-Welsch (§5.16). Our contribution is the calibrated Bayesian posterior, not the only way to robustify Phase 1 and 2.

F.3 Calibration and inference

Q8: Why doesn't trace-η give nominal coverage at 20% contamination? Single-cross-fit single-functional η_{tr} gives 83% [Wilson 66%, 93%] at 20% density — close but under-nominal because of a small constant bias (§5.7). Modular-Bayes pooling at $M = 8$ restores 100% [Wilson 89%, 100%] (§5.12). RBCI ω-tuning at $\omega = 2$ also recovers nominal coverage at the cost of width (§5.16). The trace formula is the sharper but slightly under-nominal default; both alternatives are exposed.

Q9: How does $\widehat{\eta}$ behave at higher p? At $p = 50$, $\widehat{I}$ becomes indefinite on average and $\widehat{\eta}$ accumulates substantial bias (§5.16). For $p \geq 50$ we recommend modular-Bayes pooling rather than trace-formula η.

Q10: Does nuisance uncertainty propagate? Not in single-cross-fit posteriors. Modular-Bayes pooling (§4.4, §5.12) draws M Bayesian-bootstrap nuisance fits, runs Phase-3 NUTS for each, and pools by concatenation or Rubin's rules. At 30 seeds and $M = 8$, modular pooling restores nominal ATE coverage at every density tested.

F.4 Practical configuration

Q11: What `contamination_severity` should I use? `"none"` on truly clean data (Hill index $\hat{\alpha} > 5$), `"severe"` when contamination is plausible ($\hat{\alpha} \leq 2$), `"mild"` or `"moderate"` between. Auto-detection from a Hill estimator on residuals (§5.14) works at light contamination (5%) but fails at heavy contamination (20%) because the non-robust pre-fit is itself contaminated. Domain knowledge is the safer route. See §6.9 for the full configuration tree.

Q12: When should I enable `normalize_y_for_nuisance`? On dollar-scale or other extreme-scale outcomes. The minimax-δ prescription assumes standardised residuals; raw-scale outcomes miscalibrate δ. The Lalonde NSW result (§5.9) and the scale-sensitivity probe (§6.3) confirm modest bias without pre-standardisation.

Q13: When do I need overlap weights? When $\min \hat{\pi}(X) < 0.05$ or $\max \hat{\pi}(X) > 0.95$. The library auto-warns and auto-enables at $\hat{\pi} \notin [0.02, 0.98]$ (round-13 fallback). Empirical validation: §5.10 shows overlap weights are necessary but not sufficient under combined low-overlap + contamination.

Q14: When do I need modular Bayes? Compute the dispersion ratio $\rho = \text{std}_K(\bar{\beta}^{(k)})/\text{mean}_K(\text{CI width}^{(k)})$ across $K = 5$ re-cross-fits (§5.7). $\rho > 0.15$ indicates cross-fit instability; enable modular Bayes with $M = 8$. Severity=severe consistently has $\rho \leq 0.12$ and generally does not need modular pooling at the cost of typical intervals.

F.5 Empirical scope

Q15: Why is IHDP not statistically significant? At 5 reps no method differs significantly from RX-Learner at $\alpha = 0.05$; at 10 reps the rank ordering changes and the abstract claim is "competitive, not dominant" (§5.2). IHDP is a clean-data benchmark where robust methods are not expected to win.

Q16: Have you tested non-whale contamination? Yes: Pareto ($\alpha = 1.5$), Student-t_2, Student-t_3, Student-t_5, asymmetric (treated-only), bimodal (sign-symmetric), and α-stable (round-13). All handled by `severity="severe"` with bias ≤ 0.13.

Q17: Have you tested real heavy-tailed data? Yes: Hillstrom RCT ($N = 42{,}613$, Hill index $\hat{\alpha} \approx 2$ on positive spend, see Figure 3) and Lalonde NSW ($N = 445$, dollar-scale earnings). Hillstrom passes placebo + symmetry + propensity-stratified placebo + holdout (§5.9).

Q18: How does the method scale? Single fit: 5−20 s for RX-Learner Phase 3 NUTS at $N = 1000$, $p \leq 100$ (§5.16). At $N = 10{,}000$, $p = 100$: 4.2 s, +53 MB peak RSS (§5.16). Modular-Bayes pooling multiplies runtime by $M \in [5, 10]$.

F.6 Comparison to alternatives

Q19: How does the Welsch posterior compare to RBCI ω-tuned squared-loss generalised Bayes? RBCI ω-tuning of squared-loss (round-13, §5.7) selects ω to match a bootstrap variance target; it gives wider intervals than Welsch + trace-η at comparable coverage, with bias scaling like the non-robust ATE under contamination. The Welsch redescender adds *bias control* on top of the temperature-only calibration that ω-tuning provides.

Q20: What about Lasso-DR, Catoni-DR, sandwich-CI DR-learners? Round-13 (§5.16) tests all three. Lasso-DR (sparse linear regression on DR pseudo-outcomes) and Catoni-DR (heavy-tail-robust mean estimator) are unbounded-influence and fail just like Huber-DR under contamination. Tail-trimmed-IPW (round-7) catastrophically fails because trimming changes the estimand. Bounded-influence at the regression / likelihood layer is the operational requirement.

Q21: Why not GRF (Generalised Random Forests)? GRF is a strong nonparametric CATE alternative for clean / mildly contaminated data. We did not run it because the reference implementation is in R and we kept the pipeline single-language (see §6.7). A multi-language reproduction is a natural extension.

F.7 Software and reproducibility

Q22: Can the library detect and warn about edge cases? Yes. The current implementation warns when (i) min/max $\hat{\pi}$ is outside $[0.05, 0.95]$ (small-eigenvalue risk) and (ii) auto-enables `use_overlap=True` when $\hat{\pi}$ falls outside $[0.02, 0.98]$ (round-13 auto-fallback). Users can set `use_overlap` explicitly to silence the fallback.

Q23: Are there regression tests? Yes — `tests/test_default_config_regression.py` pins the default configuration and the `contamination_severity` API; 9 tests pass on every commit. The library is configured for deterministic runs given a `random_state`.